

Projective Dynamic Logo (PDL)

Global Mapping of Structures, Results, and Open Problems

Version 28 — A Guide for Continuation of the Programme

Cédric Laubscher

Independent researcher

cedriclaubscher.ch · zenodo.org/communities/pdl-framework

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Abstract

This document provides a systematic and self-contained guide to the Projective Dynamic Logo (PDL) programme, intended for a scientific colleague who wishes to understand, verify, or continue the research. PDL is a foundational physics programme that derives fundamental physical constants and dynamical laws from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields. The minimal admissible closure under the axioms is the complete signed graph K_4 on four vertices and six edges, identified with the electron prototype. The proton is characterised by a unique integer quintuplet (24, 28, 930, 10087, 11017), from which all results flow.

The current corpus (D01–D61, plus auxiliary documents, the PDL-V series DL01–DL02, and the joint PDL–OFN bridge note N01) has reached the following state:

- Three critical Gates have been resolved: the axiomatic derivation of the amplitude 155/11017 (Gate 1, D29), the structural proof that the QCD correction coefficient equals 2 (Gate 2, D30), and the theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ (Gate 3, D31/D36).
- The quantum dynamics layer is **complete**: the Schrödinger equation (D32), Dirac equation (D33), Born’s rule Level 1 via Gleason uniqueness (D34), the Einstein coupling equation (D35), Born’s rule Level 2 including U(1) phase freedom (D46), and the London equation (D49) have all been derived from C1–C4 with no free parameter.
- Newton’s gravitational constant G is a theorem, modulo a single irreducible external parameter ($\Delta m_{\text{iso}} = m_d - m_u$).
- The Hubble tension is resolved at 0.006% with zero free parameters (D35/D27).
- Black hole thermodynamics is **complete**: the area law $S \propto R_{\text{surf}}$ (D37), the Bekenstein–Hawking formula $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ (D38), the one-quarter coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A/(4\ell_P^2)$ (D50), and the London equation (D49) are all unconditional theorems.
- Open Problem OP1 has been fully resolved by D42: the Indifference Lemma H3 is now an unconditional theorem of axioms C1–C4.
- D46 resolves OP4: the U(1) phase freedom of quantum mechanics is derived from the K_4 pulsation structure, completing the quantum dynamics layer.
- D47 resolves OP13 and OP14: the spin–orbit splitting $\Delta n = 4$ is forced by a unique perfect-square discriminant (149^2), the sub-shell filling rates $r_{\text{exc}}(Z)$ are derived analytically for all $Z = 1\text{--}82$, the Mirror Lemma establishes proton–neutron HO-PDL level isomorphism, and the periodic table (valley of stability) is a corollary of C1–C4. The nuclear stability layer is **complete**.
- D48v3 fully resolves OP2-D35 and OP-spin: the coherence stress-energy tensor C_{coh} is derived with four unconditional components. $C^{(\text{mass})}$ and $C^{(\text{orb})}$ are unconditional theorems; $C^{(\text{spin})} = 0$ in the homogeneous ground state (theorem from explicit computation over all 8 coherent configurations of K_4 , resolving OP-spin); $C^{(\text{leak})}$ is a theorem via D51+D52 in dimensionally consistent units J m^{-3} . Four dimensional errors from v1/v2 are corrected.

- D49 resolves OP-London: the London equation $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ is derived from C1–C4 via the metric identity $\nu = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$ (768/768 exhaustive verification) and axiom C4.
- D50 resolves OP12/BH-3: the coefficient $\frac{1}{4}$ in the Bekenstein–Hawking formula is derived from C1–C4 as the stable fraction 4/16 per surface relation, under H3 (D42).
- D51 and D52 together resolve OP1-D35: the cosmological leakage constant is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational bound to 0.17 ppm. The causal chain $\text{C1–C4} \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$ is complete.
- D53 provides a self-contained synthesis of the complete causal chain $\text{C1–C4} \rightarrow \Lambda_{\text{PDL}}$, with an independent numerical verification of C_{PDL} to 0.41 ppm relative to D51, and a systematic robustness analysis of the prime exponent structure. The chain spans fifty-seven orders of magnitude from combinatorial axioms to the cosmological constant.
- D40 derives the complete architecture of nuclear stability from the proton and neutron quintuplets alone: $N_{\min}(Z) = Z$ for $Z \leq 20$ as an exact theorem, the conflict saturation $C(Z > 20) = 190 T_{pp}$ exactly, and the valley of stability reproduced at 100% accuracy for all $Z = 1$ to 82.
- D41 confronts the PDL framework with the experimental results of Ha et al. (*Nature Communications* 16, 10631, 2025), confirming the structural identity $T \approx (\Delta n + 1)^2 = 25$ across three independent observables, with the wave-function component ratio agreeing at the 2% level.
- D43 establishes the computational infrastructure of the programme and proves the geometric leakage parameter $\varepsilon_{\text{geom}}(p) = 329/10087$ as an unconditional theorem (OP-A resolved).
- D44 derives the hierarchical filter factor $k = 0.921716$ from C1–C4 as an unconditional theorem (OP-B resolved). The complete causal chain $\text{C1–C4} \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G$ is now fully closed with no remaining open problem and no free parameter beyond Δm_{iso} .
- D45 derives the first falsifiable astrophysical confrontation: the primordial black hole survival threshold is shifted upward by +11.89% relative to the standard GR value, giving $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$, testable with existing Fermi-LAT data via BLACKHAWK and ISATIS.
- **New in v22 — D54** resolves OP-pressure [1]: the equation of state of the PDL coherence fluid is $w(N) = -\rho_\Lambda / \rho_{\text{coh}}(N)$, an unconditional theorem of C1–C4. In the homogeneous ground state, $w_{\text{mass}} = w_{\text{spin}} = w_{\text{orb}} = 0$ exactly (three distinct theorems), and $w_{\text{leak}} = -1$ intrinsically. The PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = \rho_\Lambda / \rho_{\text{crit}} = 0.6838$ agrees with the Planck 2020 value $\Omega_\Lambda^{\text{obs}} = 0.685$ to 0.17%, with no cosmological parameter as input. The PDL coherence fluid interpolates between pressureless cold matter at sub-nuclear scales and dark energy at cosmological scales — both emerging from the same combinatorial structure with no parameter adjustment.
- **New in v23 — D55** resolves OP10 at the level of the Weinberg angle [2]: the formula $\theta_W = 19\pi/119$ is derived as an unconditional theorem of C1–C4, via four established results (D33, D46, D51, and the new Lemma D) and zero free parameters. The key structural result is that θ_W is a dephasing sequencing ratio: the Berry phase accumulated during $k_2 = 19$ elementary interface steps out of $N_{\text{total}} = 119$, where $k_2 = n_u - R_e + 1$ (D51 theorem) and $N_{\text{total}} = (n_d/4) \cdot (r_d - r_u)/R_e = 7 \times 17$ (Lemma D, proved exhaustively over all 768 $(A) \wedge (B)$ configurations). The predicted $\sin^2 \theta_W = 0.231196$ agrees with the PDG 2024 value 0.23121 ± 0.00003 at 0.48σ . As a secondary result, combining with the geometric zeroth-order formula yields a new PDL prediction $\Delta m_{\text{iso}} = 2.446 \text{ MeV}$ (within 0.92σ of FLAG 2024), falsifiable by next-generation lattice QCD. The electroweak sector of the Standard Model is now structurally derived from C1–C4.
- **New in v24 — D56** resolves OP-D41-1-A [3]: the number of maximal partial closures on a k p- k h nuclear interface surface is $N_{\text{comp}}(k) = k$ exactly, proved from three lemmas (uniqueness on one unit, disjointness of mixed-triangle sets, C3-irreducibility of the assembly) and verified for $k = 1, \dots, 5$.
- **New in v28 — D61** resolves OP-D59-2 [4]: the minimal covariant derivative $D_\mu = \partial_\mu - ig A_\mu$ is derived from C4 as a survival theorem. The group of C4-admissible discrete transports on K_4 is $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ (2048 exhaustive checks); V_4 acts faithfully on this group by automorphisms

with fixed-point subgroup $F \cong V_4$ labelled by the three perfect matchings of K_4 (288 checks); the equivariant connected completion is $G_{\text{gauge}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, consistent with D46–D60; and Utiyama’s uniqueness theorem identifies D_μ as the unique first-order equivariant generator. The gauge sector of the Standard Model — group, representations, and gauge dynamics — is now completely derived from C1–C4, without variational postulate and without free parameter at the structural level.

- **New in v28 — D60** [5] proves Hypothesis $\text{H}_{\text{SU}2}$ as an unconditional theorem of C1 and C2 alone (OP-D57-1 resolved). The V_4 -invariance of the pulsation structure forces $G_{\text{eff}} = S_4/V_4 \cong S_3$, establishing the complete logical chain from the axiom of existence (C1) to the effective gauge symmetry group, with no conjectural step remaining in the gauge sector.
- **New in v25 — D57** [6] provides the first combinatorial derivation of the tree-level Weinberg angle from C1–C4 alone: $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$, where $R_e = 6$ is the relational budget of K_4 (number of edges), itself a theorem of C1–C4 (D16a). Two independent routes are proved equivalent: the combinatorial route ($192/768 = 1/4$ from the $(A) \wedge (B)$ criterion, D29) and the orbit route ($\dim(1)/\dim(\text{orbit-4}) = 1/4$ from the S_4 action on coherent configurations). Both routes identify the factor $1/4$ with $|V_4|/16$. The quotient $A_4/V_4 \cong \mathbb{Z}_3$ is identified as the combinatorial origin of $\beta_1(K_4) = 3$, providing a partial resolution of OP-OFN-1 (N01 [7]).
- **New in v25 — N01** (Laubscher & Evdokimov) establishes the PDL–OFN bridge [7]: $\beta_1(K_4) = 3$ and $b_1(\Omega_{21}) = 3$ under Hamming distance-1 adjacency are the same topological invariant viewed from two independent frameworks. Five Python scripts verified the convergence exhaustively. Open Problem OP-OFN-1 (formal derivation of the link between the three PDL leakage cycles and the three OFN fermion generations) is identified; partial resolution provided by $A_4/V_4 \cong \mathbb{Z}_3$ (D57).
- **New in v22 — PDL-V programme (DL series)**: Documents DL01 [8] and DL02 [9] inaugurate a structurally distinct extension of the PDL framework to the threshold conditions for life and consciousness. DL01 introduces the self-replication operator S and establishes that $S(K_4) \cong K_4$ (8/8 coherent configurations), that $\delta^* = 1/2$ is an invariant, and that the rarity law $f(n) = 2^{n+1} - \binom{n}{2}$ holds exactly. DL02 proves three analytical theorems: K_4 is the unique fixed point of S (Theorem 1); the life threshold strictly precedes the consciousness threshold, $n_{\text{life}}^* < n_{\text{consciousness}}^*$ (Theorem 2); and both thresholds are finite, $n_{\text{max}}^* < \infty$ (Theorem 3), giving the hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$ as an unconditional theorem of C1–C4. These documents are presented in Part III of this mapping, clearly separated from the physical constants programme of Part I and II.
- **New in v27 — D59** constructs the fundamental representation **3** of $\text{SU}(3)$ from C1–C4 and identifies the physical colour gauge group $\text{SU}(3)_c$. The carrier space is the trace-zero plane $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$, whose axes are canonically labelled by $V_4 \setminus \{e\}$. Five lemmas verified by exhaustive exact-integer computation (PDL_D59_script1.py [10]): (L1) S_3 -action on $V_4 \setminus \{e\}$ faithful and transitive; (L2) $\chi_\pi = \chi_{\text{trivial}} + \chi_{\text{standard}}$; (L3) $W = \{a + b + c = 0\}$ unique S_3 -invariant complement of $\mathbb{C}(1, 1, 1)$; (L4) A_2 root system in W , Cartan matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$; (L5) isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ (D47, theorem of C4) selects **3** over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms. Exact identity: $k_3/(k_1 + k_2) = R_e = 6$. Negative result documented: bijection $(T_2) \leftrightarrow (T_3)$ physically motivated but not S_4 -equivariant. OP-D58-1 and OP-D58-3 resolved.

There is no longer any foundational open problem at the axiomatic level. The causal chain from C1–C4 to the cosmological constant Λ_{PDL} is complete. The equation of state of the coherence fluid (D54) is fully resolved. The Weinberg angle (D55) is derived as an unconditional theorem, closing OP10 at the level of θ_W . The remaining open problems concern higher-level derivations (electroweak W/Z mass ratio, quark mass spectrum, three fermion generations), extensions (superheavy nuclei, dark matter identification), and experimental confrontations. A new structural hypothesis OP-D59-2 is identified: the minimal covariant derivative $D_\mu = \partial_\mu - igA_\mu$ may be the unique C4-compatible gauge transport, making gauge dynamics a consequence of the survival principle.

Contents

1	Purpose and Scope	6
1.1	Aim of this document	6
1.2	What makes this version different from v26	6
1.3	Epistemic conventions	6
2	Document Corpus	7
2.1	Complete document table	7
2.2	Reading guide for a new collaborator	15
3	Axiomatic Foundations	16
3.1	The four relational axioms C1–C4	16
3.2	The minimal admissible closure: K_4	16
3.3	The proton quintuplet	16
3.4	The primary leakage parameter κ	17
3.5	The proton–electron mass ratio μ	17
4	The Three Gates: Critical Proofs	17
4.1	Gate 1: Axiomatic derivation of 155/11017	17
4.2	Gate 2: The QCD correction coefficient $a = 2$	17
4.3	Gate 3: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$	17
5	Derived Dynamical Equations	18
5.1	The Schrödinger equation (D32)	18
5.2	The Dirac equation (D33)	18
5.3	Born’s rule (D34 and D46)	18
5.4	The Einstein coupling equation (D35 and D48)	18
6	Black Hole Thermodynamics	18
6.1	The area law as an unconditional theorem (D37)	18
6.2	The Bekenstein–Hawking formula (D38)	19
7	The Coherence Stress-Energy Tensor (D48)	19
7.1	Double determination of the coherence volume	19
7.2	The coherence mass density	19
7.3	Components of the coherence stress-energy tensor	19
8	Equation of State of the Coherence Fluid (D54)	20
9	Nuclear Stability and the Periodic Table	20
9.1	The neutron survival condition (D40)	20
9.2	Conflict saturation (D40)	20
9.3	Spin–orbit splitting from axioms (D47)	21
9.4	Sub-shell filling rates and the periodic table (D47)	21
9.5	Nuclear spectroscopy confrontation (D41) and the closure multiplicity theorem (D56)	21
10	Cosmology	22
10.1	The density function $\sigma(N)(N)$	22
10.2	The Hubble tension resolution	22
11	The PDL–QCD Interface	22
12	Causal Chain Closure and Geometric Leakage (D43–D44)	22

13	Epistemic Architecture	23
14	Open Problems	25
15	Logical Dependency Map	30
16	Logical Dependency Map	30
17	Falsifiable Predictions	32
18	Continuation Guide	33
18.1	Entry points	33
18.2	Recommended next documents	33
19	Glossary	34
I	The PDL-V Programme: Life and Consciousness	36
20	Introduction to the PDL-V Programme	36
20.1	Separation from the physics programme	36
20.2	Reading note	36
21	Self-Replication Operators and the Rarity Law (DL01)	36
22	Threshold Hierarchy Theorems (DL02)	36
22.1	Open problems in the PDL-V programme	37

1 Purpose and Scope

1.1 Aim of this document

This mapping document pursues four objectives. First, it provides a complete and current inventory of the PDL corpus (D01–D48, plus auxiliary documents), with concise descriptions of each document’s role and main result. Second, it makes explicit the logical dependency structure of the programme: which results follow from which, what has been proved, and what remains conjectural or open. Third, it classifies all results according to a rigorous epistemic taxonomy, distinguishing proved theorems, conditional theorems whose hypotheses are explicitly named, structurally motivated conjectures, and open problems. Fourth, it identifies the most productive entry points and continuation paths for a colleague wishing to extend the programme. This document is not a replacement for the primary technical documents; it is a navigational and epistemic guide that complements them.

1.2 What makes this version different from v26

Version 26 covered the corpus up to D58 and established $SU(3) \times SU(2) \times U(1)$ as an unconditional algebraic theorem of C1–C4. The present version v27 adds Document D59 and its supplementary verification script `PDL_D59_script1.py`, resolving two open problems of D58 and completing the first derivation of a gauge representation from purely combinatorial axioms.

- **New in v27 — D59** constructs the fundamental representation **3** of $SU(3)$ from C1–C4 and identifies the physical colour gauge group $SU(3)_c$. The carrier space is the trace-zero plane $W = \{(a, b, c) \in \mathbb{C}^3 : a + b + c = 0\}$, with axes canonically labelled by $V_4 \setminus \{e\}$ (D58, Lemma L2). Five lemmas verified by exhaustive exact-integer arithmetic in `PDL_D59_script1.py` [10]: (L1) $S_3 = S_4/V_4$ acts faithfully and transitively on $V_4 \setminus \{e\}$ (recalled from D58); (L2) character decomposition $\chi_\pi = \chi_{\text{trivial}} + \chi_{\text{standard}}$, verified over all 24 elements of S_4 ; (L3) $W = \{a+b+c=0\}$ is the unique S_3 -invariant complement of $\mathbb{C}(1, 1, 1)$ in $\mathbb{C}^3$; (L4) A_2 root system in W with Cartan matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, confirmed by exact integer inner products; (L5) the isospin asymmetry $\Delta n = n_d - n_u = 4 > 0$ (unconditional theorem of C4, D47) forces the canonical orientation selecting **3** over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms. Exact identity: $k_3/(k_1 + k_2) = R_e = 6$.
- A documented negative result shows that the bijection between the leakage-cycle triplet $(T_2) = \{9, 19, 168\}$ and the orbit triplet $(T_3) = V_4 \setminus \{e\}$ is physically motivated by the quintuplet but not S_4 -equivariant.
- Open problems resolved by D59: OP-D58-1 (physical identification of $SU(3)_c$) and OP-D58-3 (fundamental representation **3** from C1–C4).
- New structural hypothesis identified: the minimal covariant derivative $D_\mu = \partial_\mu - igA_\mu$ is the unique gauge-transport structure compatible with C4 (any non-equivariant transport introduces a cost eliminated by C4). Stated as OP-D59-2.

The reader familiar with v26 should note: D59-py and D59 added to corpus table; OP-D58-1 and OP-D58-3 marked resolved; OP-D59-1 (quark mass spectrum) and OP-D59-2 (minimal covariant derivative from C4) added; dependency diagram updated; Recommended next documents updated; bibliography gains two entries (D59py, D59).

1.3 Epistemic conventions

Throughout this document, the following colour-coded epistemic labels are used in tables and text:

Theorem A result established by a complete proof from the PDL axioms C1–C4 and previously proved theorems, with no free parameters and no unproved auxiliary hypotheses.

Theorem (Hx)

A result proved under one or more explicitly named hypotheses $H1, H2, H3, \dots$; the proof itself is complete conditional on those hypotheses.

Conjecture

A structurally motivated claim supported by numerical evidence, qualitative arguments, or partial proofs, but not yet established rigorously.

Named Hypothesis

An auxiliary assumption that is explicitly labelled and isolated, so that results depending on it are clearly traceable.

Open Problem

A well-posed question whose resolution would advance the programme significantly.

2 Document Corpus

2.1 Complete document table

Table 1 lists the complete PDL corpus in Zenodo canonical order.

Table 1 PDL corpus in Zenodo canonical order. *Main result* denotes the single most consequential contribution of each document to the programme chain.

Label	Title (abbreviated)	Main result / role
D01	The Emergence of Physical Reality within the PDL Framework [11]	Founding article: four relational axioms, structure, proton architecture, first treatment of α, μ, G , and cosmology.
D02	Introduction to the PDL [12]	High-level conceptual overview; standard first reading for new collaborators.
D01F	L'émergence de la réalité physique (LDP FR) [13]	French translation of D01.
D03	The Emergence of Physical Reality (IMRaD format) [14]	Journal-formatted version of D01.
D04	A Modal Dialogue between PDL and the Theory of Objectivity [15]	Comparative study of PDL and TO; ontological commitments of PDL.
D05	A Minimal Relational Sketch for the Golden Ratio in PDL [16]	Self-similar core–surface partition; golden ratio and active surface $(\sigma, 3)$.
D06	Hierarchical Filtering of Coherence Leakage and the Exponent 18 [17]	First formulation of the 18-filter leakage hierarchy.
D07	Gleason-Type Uniqueness of Born's Rule for Spin- $\frac{1}{2}$ [18]	Early sketch of the Gleason approach; superseded by D34 at Level 1.
D08	Logical Leakage as Self-Maintained Probability: Topological Reformulation [19]	Coexistence topology; coherence-cost pseudo-metric; leakage as probability.

Label	Title (abbreviated)	Main result / role
D09	PDL as a Foundational Research Programme [20]	Programme position paper; first systematic open-problem list.
D10	From Discrete Coherence Flux to Effective Fields [21]	Gauss- and Faraday-like relations from coherence fluxes; path towards Schrödinger.
D10a	Proper Time as Coherence-Cycle Counting [22]	Proper time as a count of coherence cycles; emergent Minkowski structure; $\omega_p = m_p c^2 / \hbar$ forces $\lambda_C = \hbar / (m_p c)$ and $V_C = (4/3)\pi\lambda_C^3$ (chain 1 confirmed in D48v3; chain 2 is a geometric consistency check, not an independent proof — see D48v3 Remark 3.1).
D11	Towards an Einstein–Dirac Unification from PDL [23]	Preliminary sketch coupling Dirac spinors to the emergent coherence-cost metric.
D12	A Structural Derivation of the Fine-Structure Constant from PDL [24]	Closed-form expression for α from $(930, \sigma, \varphi)$; accuracy $\mathcal{O}(10^{-4})$.
D13	Compatibility of the Schrödinger Framework with PDL [25]	Benchmark: PDL-derived α and μ in hydrogenic Schrödinger; spectral concordance.
D14	Born’s Rule and the Golden-Ratio Active Surface in PDL [26]	Relational derivation of Born’s rule for spin- $\frac{1}{2}$; role of φ in σ .
D15	Towards a Schrödinger-Type Dynamics from PDL: A Relational Sketch [27]	Early sketch of Schrödinger emergence; superseded by D32.
D16	Combinatorial Selection of the Proton Architecture: Uniqueness Conjecture [28]	Optimisation functional for the quintuplet; local uniqueness conjecture.
D16a	Minimal Stationary Closures in PDL: Necessity of the $(4, 6)$ Block [29]	Theorem: no admissible closure exists on $ V < 3$; K_4 is first admissible.
D16b	Combinatorial Selection and Local Uniqueness of the Proton Architecture [30]	Refined combinatorial treatment; local uniqueness conjecture for $(24, 28, 930, 10087, 11017)$.
D17	Coherence Leakage, Hierarchical Filtering, and the Exponent 18 [31]	Organisation of the 18 filters into three blocks; effective coupling ansatz ε^{18} .
D18	Discrete Cavity Modes and Logical Density of States [32]	PDL cavity modes; $2n^2$ density-of-states in three dimensions.
D19	Existence as Pulsating Closure: An Ontological Meditation [33]	Philosophical analysis of PDL boundaries, leakage, and transcendence.
D20F	Qui que nous puissions être (FR) [34]	French philosophical synthesis; companion to D20.

Label	Title (abbreviated)	Main result / role
D20	Whoever We May Be: A Philosophical Synthesis of PDL [35]	Ontological and existential synthesis; accessible entry point for non-specialists.
D21	Universal Coherence Leakage: A Structural Bridge between G and α [36]	Primary result: $R_{\text{tot}}/R_{\text{surf}} = 2\nu^2/(2\nu^2 - 1)$; $G \approx 6.67448 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ (27 ppm vs. CODATA 2022).
DN	<i>Whatever We May Be</i> : The PDL popular book [37]	Popular science monograph; non-technical introduction to PDL.
D22	Nuclear Stability and the Periodic Table as Combinatorial Closure Hierarchies [38]	Neutron quintuplet (24, 28, 1032, 9960, 10992); $N_{\text{crit,max}} = 126.1$; $Z_{\text{sat}} = 20$; Mirror Lemma (confirmed analytically in D47).
DM	PDL Global Mapping v20 [39]	Navigation document for the complete corpus D01–D53; the present document.
D23	Topological Origin of the Exponent 18, S^2 , and the Periodic Table [40]	Theorem: $18 = 6 + 5 + 4 + 3$ is topologically necessary via S^2 ; $n = 3$ forced by S^2 topology (consistency with D48v3; the geometric argument is not an independent chain for V_C — see D48v3 Remark 3.1).
D24	Closure-Density Dependence of κ and the Hubble Tension [41]	Introduction of $\sigma(N) = 1 - (1 - \kappa)^N$; structural origin of H_0 discrepancy.
D25	A Parameter-Free Structural Bridge between α and G [42]	Primary result: full bridge formula; G at 27 ppm; $\kappa \approx 0.0075194$.
D26	Cosmological Resolution via PDL: G as Topology-Dependent [43]	G as topology-dependent parameter; Hubble tension as scale-transition signature.
D27	Structural Derivation of N_{CMB} from the PDL Neutron Architecture [44]	$N_{\text{CMB}} = 40$ from n_6 ; $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$; 0.27σ Planck.
D28	The PDL–QCD Boundary: Mass Ratio, Correction, and the ε_G^B Conjecture [45]	$\mu = \frac{155}{11017}(1 - 2\kappa^2) \frac{m_{\text{iso}}}{m_p}$; ε_G^B conjecture formulated.
D29	Axiomatic Derivation of 155/11017: Gate 1 Resolved [46]	Theorem: 155/11017 proved from C1–C4 via AB; 768/768 exhaustive verification.
D30	QCD Correction Coefficient $a = 2$: Gate 2 Resolved [47]	Theorem: $a = 2$ structurally forced; $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter.
D31	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$: Gate 3 (preliminary) [48]	First proof of Gate 3 under linear bridge inversion hypothesis; superseded in rigour by D36.

Label	Title (abbreviated)	Main result / role
D32	Schrödinger Dynamics from the AB Coupling Criterion [49]	Theorem: Schrödinger equation derived; $b = -i\hbar/(2m_e)$; ∂_x^2 forced by AB.
D33	Dirac Equation from the $SL(2, \mathbb{C})$ Pulsation of K_4 [50]	Theorem: $T = -i\sigma_2$ unique among 8 candidates; $T^2 = -I_2$; Clifford algebra; NR limit = D32 exactly.
D34	Born Rule from AB-Admissible Amplitudes [51]	Theorem Level 1: five Gleason axioms verified; Born rule unique from Gleason uniqueness.
D35	Quantitative Einstein Equation from $\sigma(N)$ [52]	Theorem (now unconditional via D42): $G_{\text{eff}}(N) = G_{\text{Newton}}$; Hubble ratio at 0.006%; synthesis document.
D36	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ from the Trace Structure of K_4 [53]	Theorem (now unconditional via D42): $\sigma_0 = 0$ exact from AB; $\text{Tr}(A) = 0$ proved; Gate 3 strengthened.
D37	Surface Locality of Relational Entropy and the PDL Area Law [54]	Theorem BH-1: entropy of a PDL closure is entirely carried by R_{surf} ; area law $S \propto R_{\text{surf}}$ unconditional.
D38	Bekenstein–Hawking Entropy from the PDL Effective Gravitational Coupling [55]	Theorem BH-2 (now unconditional via D42): $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$; verified to 10^{-15} ; three falsifiable PBH predictions.
D39	Derivation of κ from the PDL Axioms: the Indifference Lemma [56]	Theorem (H3): $\kappa = 3/\sigma$ under H3; partial resolution of OP1; D36 $\Rightarrow$ H3 is the unique residual; H3 now a theorem (D42).
D40	Nuclear Stability and the Periodic Table from PDL Combinatorial Axioms [57]	Theorems: $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ exact; valley of stability at 100% for $Z = 1\text{--}82$; eight assembly rules; connection identified between OP14 and OP12.
D41	Structural Closure Multiplicity and the Isospin-Symmetric Island of Inversion [58]	First PDL confrontation with nuclear spectroscopy data (Ha et al. 2025). Three independent observables consistent with $T = (\Delta n + 1)^2 = 25$: BE2 ratio (22%); N_{comp} ratio (2%); ΔS_{WF} (25%). Conjecture H_B introduced; falsifiable predictions P7 and P8.
D42	Derivation of H3 from Axioms C1–C4: The Equiparticipation Lemma [59]	Theorem: H3 derived unconditionally via D42-L1 (Equiparticipation), D42-L2 (cross-triangle blindness), D42-L3 (S_4 -equivariance; 24,576 cases). Corollary: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional. OP1 fully resolved.
D43	Numerical Verification of the PDL Causal Chain: 18-Filter Architecture and Colab Infrastructure [60]	Systematic numerical verification at 50-decimal precision. Proves $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems (OP-A resolved). Reusable Colab infrastructure.

Label	Title (abbreviated)	Main result / role
D44	Closure of OP-B: Derivation of the Hierarchical Filter Factor k from the PDL Axioms [61]	Theorem: $k = 0.921716$ derived from C1–C4 unconditionally. OP-B resolved; OP8 resolved; complete causal chain C1–C4 $\rightarrow G$ closed.
D45	Primordial Black Hole Evaporation Threshold from PDL: A Falsifiable Prediction for Fermi-LAT [62]	First falsifiable astrophysical confrontation. $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$ (+11.89%). Excess Hawking flux at $E_{\text{peak}} \approx 90\text{--}104 \text{ MeV}$ in IGRB; test protocol via BLACKHAWK + ISATIS + Fermi-LAT.
D46	Born’s Rule Level 2: Derivation of U(1) Phase Freedom from the K_4 Pulsation [63]	Theorem OP4 resolved: Proposition 1 — K_4 sign equivalence $s \sim -s$ and $T^2 = -I_2$ generate a canonical U(1) action with Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. Proposition 2 — $(A) \wedge (B)$ criterion is U(1)-invariant (verified over all 8 configurations). Proposition 3 — unique U(1)-equivariant measure on S^3 satisfying Level 1 Gleason axioms is Born’s rule. Quantum dynamics layer complete. Connection to Berry phase and London/Meissner analogy via $C^{(\text{orb})}$ (D48) documented.
D47	Derivation of Sub-Shell Filling Rates and the Periodic Table from the PDL Axioms: Resolution of OP13 and OP14 [64]	Theorem OP13 resolved: $\Delta n = 4$ is the unique value in $\{0, 4, 8, \dots\}$ such that the discriminant of the quasi-completeness equation is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ unconditionally. Mirror Lemma (Lemma 5.2): proton and neutron HO-PDL levels are isomorphic, 20 levels verified. Theorem OP14 resolved: $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise, 31/31 sub-shell boundaries, zero free parameter. Corollary: $N_{\text{min}}(Z)$ for $Z = 1\text{--}82$ is an unconditional theorem of C1–C4. Nuclear stability layer complete.
D48v3	Derivation of the Coherence Stress-Energy Tensor from the PDL Axioms: Explicit Form of C_{coh} , Resolution of OP2-D35 and OP-spin [65]	Theorem OP2-D35 and OP-spin fully resolved (D48v3, DOI 10.5281/zenodo.20151380). V_C theorem via D10a (Chain 1); $\rho_{\text{coh}}(N)$ unconditional. $C^{(\text{mass})}$ unconditional theorem. $C^{(\text{spin})} = 0$ in the ground state: theorem from $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ over all 8 K_4 configurations (Weyssenhoff–Raabe). $C^{(\text{orb})} = m_p^2 \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ (Madelung), vanishes in ground state (D49). $C^{(\text{leak})}$ in J m^{-3} via D51+D52. PDL Einstein equation dimensionally consistent. Four corrections from v1/v2 documented (Section 10 of D48v3).
D49	Derivation of the London Equation from the PDL Axioms: Resolution of OP-London [66]	Theorem OP-London resolved. Lemma: $\nu = \frac{1}{4} \ \psi_1 - T^k \psi_1\ _{\mathcal{H}_{\text{cyc}}}^2$ verified 768/768 in exact integer arithmetic. Axiom C4 forces $\psi_j = \psi_1$ over V_C (via C3 connectivity), hence $\phi = \text{const}$ (London gauge). Corollary: $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$; $\lambda_{L, \text{PDL}}(N = 40) \approx 7.25 \times 10^{-15} \text{ m}$.

Label	Title (abbreviated)	Main result / role
D50	Derivation of the Bekenstein–Hawking One-Quarter Coefficient from the PDL Axioms: Resolution of OP12/BH-3 [67]	Theorem OP12/BH-3 resolved. The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ equals the stable fraction $4/16 = 1/4$ per surface relation, proved from D29 (exhaustive, 768/768) and H3 (D42). Statistical independence $P(e_1 \wedge e_2) = 1/16 = (1/4)^2$ verified over 30 720 cases. $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$; $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352$ nats. Black hole thermodynamics layer complete.
D51	The Cosmological Leakage Constant from the PDL Axioms: A Structural Derivation of C and Resolution of OP1-D35 [68]	Theorem OP1-D35 structurally resolved. $\beta_1(K_4) = 3$ gives three leakage cycles (topology). C1+C3 forces prime cycle lengths. Indices $k_1 = n_d - n_u + R_e - 1 = 9$, $k_2 = n_u - R_e + 1 = 19$, $k_3 = R_e \cdot n_d = 168$ are exact theorems; $k_1 + k_2 = n_d$. Conjecture PDL-C: $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm of Λ_{obs} . Uniqueness verified by exhaustive exclusion of all alternative bases.
D52	Formal Identification of the Three Leakage Bases as Natural Representatives of the Three Cycles of $\beta_1(K_4) = 3$: Completion of Conjecture PDL-C [69]	Theorem OP1-D35 fully resolved. The three bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ are proved to be the natural and unique representatives of the three leakage cycles: $(1 - \kappa)$ from D42 (surface sector), $R_{\text{val}}/R_{\text{tot}}$ from D29 Lemma 2 ($\Omega_{\text{val}} = 1$, valence sector), $(1 - \eta_L)$ from D17/D30 (coupling sector). Algebraic independence verified (40 401 cases). Full causal chain: $\text{C1–C4} \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$. Cosmological leakage layer complete.
D53	From Four Axioms to the Cosmological Constant: Causal Closure of the Chain C1–C4 to Λ — A Synthesis and Computational Verification [70]	Synthesis and verification of the complete chain $\text{C1–C4} \rightarrow \Lambda_{\text{PDL}}$ in a single self-contained document. Independent numerical verification at 50-decimal precision: C_{PDL} confirmed to 0.41 ppm relative to D51; prime robustness confirmed (composite substitution degrades agreement by ≥ 4 orders of magnitude); metrological origin of the apparent 8.17 ppm offset identified (8.6 ppm in $(m_p c / \hbar)^2$ between CODATA versions, no structural significance). The chain spans 57 orders of magnitude.

Label	Title (abbreviated)	Main result / role
D54	Equation of State of the PDL Coherence Fluid: Resolution of OP-pressure [1]	Theorem OP-pressure resolved. The equation of state $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$ is an unconditional theorem of C1–C4. In the homogeneous ground state: $w_{\text{mass}} = 0$ (pressureless dust, D48v3 Thm 4.1); $w_{\text{spin}} = 0$ (D48v3 Thm 5.3, OP-spin); $w_{\text{orb}} = 0$ (D49 Thm 1 via London gauge); $w_{\text{leak}} = -1$ (cosmological constant structure, D48v3 Prop. 8.1). PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = 0.6838$ vs $\Omega_\Lambda^{\text{obs}} = 0.685$ (Planck 2020): agreement to 0.17%, no cosmological parameter as input. The PDL coherence fluid provides a unified description of pressureless cold matter (at sub-nuclear scales, $ w \sim 10^{-44}$) and dark energy (at cosmological scales) from a single combinatorial structure. Two new open problems introduced: OP-DM (dark matter identification) and OP-inhomogeneous (equation of state beyond the ground state).
D55	Derivation of the Weinberg Angle from the PDL Axioms: De-phasing Sequencing, Lemma D, and a New Prediction for Δm_{iso} [2]	Theorem OP10 partially resolved (θ_W derived). Lemma D (new unconditional theorem): the $(A) \wedge (B)$ -stable interface decomposes into $N_{\text{total}} = (n_d/4) \cdot (r_d - r_u)/R_e = 7 \times 17 = 119$ elementary steps, proved exhaustively over 768 configurations with zero counter-examples. Main result: $\theta_W = k_2\pi/119 = 19\pi/119$, where $k_2 = n_u - R_e + 1 = 19$ (D51 theorem). Berry phase identification: $\gamma_B = -\Omega/2$ with $\Omega = 4\pi k_2/119$ (D46 + Lemma D). Predicted $\sin^2 \theta_W = 0.231196$ vs PDG 2024: 0.23121 ± 0.00003 , compatible at 0.48σ . New prediction P10: $\Delta m_{\text{iso}} = 2.446$ MeV (within 0.92σ of FLAG 2024). The electroweak sector gauge geometry is structurally derived from C1–C4. Remaining open: W/Z mass ratio (OP10-c) and lattice QCD test (OP10-d).
D56	Derivation of the Linear Scaling $N_{\text{comp}}(k) = k$ from Axioms C1–C4: Resolution of OP-D41-1-A [3]	Theorem OP-D41-1-A resolved. Three lemmas: L1 (uniqueness on one unit, D16a), L2 (disjointness of mixed-triangle sets, D29), L3 (C3-irreducibility). $N_{\text{comp}}(k) = k$ verified for $k = 1, \dots, 5$.
D57	Derivation of the Tree-Level Weinberg Angle and the SU(2) Gauge Structure from C1–C4 [6]	Theorem : $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$, two independent routes (Route A: $192/768$, D29; Route B: $\dim(\mathbf{1})/\dim(\text{orbit-4})$), equivalence via $ V_4 = 4$. Conjecture H_{SU2} : C1 forces V_4 trivial $\rightarrow S_4/V_4 \cong S_3 \rightarrow \text{Dic}_3 \subset 2T \subset \text{SU}(2)$. New: $A_4/V_4 \cong \mathbb{Z}_3 \leftrightarrow \beta_1(K_4) = 3 = b_1(\Omega_{21})$ (partial resolution OP-OFN-1). D55 reinterpreted as Berry correction of π/R_e : $\theta_W = (\pi/R_e) \cdot (R_e k_2/N_{\text{tot}}) = (114/119) \cdot (\pi/6)$.

Label	Title (abbreviated)	Main result / role
N01	PDL–OFN Convergence on $\beta_1 = 3$: A Numerical Investigation (Laub-scher & Evdokimov) [7]	Five Python scripts verify exhaustively: $\beta_1(K_4) = 3$ (PDL) = $b_1(\Omega_{21}) = 3$ (OFN, Hamming dist-1 adjacency). Poses OP-OFN-1 (formal link 3 PDL cycles $\leftrightarrow$ 3 OFN generations) and OP-OFN-4 (derive $SU(3) \times SU(2) \times U(1)$ using OFN decomposition $8 \oplus 3 \oplus 1 \oplus 1$ as guide). Partial resolution of OP-OFN-1 via $A_4/V_4 \cong \mathbb{Z}_3$ (D57).
D58-py	PDL_SU3_script1.py: Verification of the Five Lemmas Establishing $SU(3)$ as an Unconditional Theorem of C1–C4 [71]	Supplementary verification script for D58. Five lemmas verified exhaustively by exact integer arithmetic (Python, numpy, networkx). DOI 10.5281/zenodo.20623231. Deposited as Software on Zenodo.
D58	Derivation of the $SU(3)$ Gauge Structure from the PDL Axioms C1–C4: Completion of the Standard Model Gauge Group [72]	Theorem (algebraic): $SU(3)$ from C1–C4 + Cartan–Killing–Weyl. Five lemmas (L1: $S_4/V_4 \cong S_3$; L2: S_4 -equivariant bijection $V_4 \setminus \{e\} \leftrightarrow V_4$ -edge-orbits; L3: $A_4/V_4 \cong \mathbb{Z}_3 = Z(SU(3))$; L4: rank-2 Cartan from $R_e = 6$; L5: A_2 root system). Negative result: V_4 acts faithfully on $H_1(K_4; \mathbb{R})$. Corollary: $SU(3) \times SU(2) \times U(1)$ algebraic theorem of C1–C4 (with D46, D57). DOI 10.5281/zenodo.20622987.
D59-py	PDL_D59_script1.py: Verification of the Five Lemmas Establishing the Fundamental Representation $\mathbf{3}$ of $SU(3)_c$ from C1–C4 [10]	Supplementary verification script for D59. Five lemmas (L1–L5) and one negative result (NEG) verified exhaustively by exact integer arithmetic (Python standard library only: <code>itertools</code> , <code>fractions</code> ; no floating-point). All 17 checks PASSED OR VERIFIED NEGATIVE. DOI 10.5281/zenodo.20628926. Deposited as Software on Zenodo.
D59	Physical Identification of $SU(3)_c$ and the Fundamental Representation $\mathbf{3}$ from the PDL Axioms C1–C4 [73]	Theorem : $SU(3)_c$ acts on the trace-zero plane $W = \{a + b + c = 0\} \subset \mathbb{C}^3$ in the fundamental representation $\mathbf{3}$, as an unconditional theorem of C1–C4 + Cartan–Killing–Weyl. Axes of W labelled by $V_4 \setminus \{e\}$ (D58, L2). Orientation $\mathbf{3}$ vs $\bar{\mathbf{3}}$ selected by $\Delta n = 4 > 0$ (D47). Exact identity $k_3/(k_1 + k_2) = R_e = 6$. Negative result: bijection $(T_2) \leftrightarrow (T_3)$ not S_4 -equivariant. Resolves OP-D58-1 and OP-D58-3. DOI 10.5281/zenodo.20629282.
— PDL-V Programme: Life and Consciousness (DL series) —		

Label	Title (abbreviated)	Main result / role
DL01	From Axioms to Life: The PDL-V Conjecture and the Hierarchical Replication of C1–C4 [8]	Computational theorems inaugurating the PDL-V programme. Lifting operator L , constraint-structure operator $S(\Gamma)$, self-representation relation $R^k(\Gamma)$ defined formally. $S(K_4) \cong K_4$: proved over 8/8 coherent configurations. R_{weak}^1 forced at 100% for all $n \geq 4$ (verified $n = 4-7$, 8/8 to 64/64). $\delta^* = 1/2$: invariant of the optimal lineage variation = half-pulsation of K_4 . Rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ with exponents equal to triangular numbers: exact theorem.
DL02	Existence, Separation, and Finiteness of the Life and Consciousness Thresholds in the PDL-V Programme [9]	Analytical theorems on the life/consciousness hierarchy. Theorem 1: K_4 is the unique fixed point of $S - C(4, 3) = 4$ is the unique positive integer solution of $(n-1)(n-2) = 6$. Theorem 2: $n_{\text{life}}^* < n_{\text{consciousness}}^*$ — structurally forced for $n \geq 5$ from $ S^1 \neq S^2 $. Theorem 3: $n_{\text{max}}^* < \infty$ — from the doubly exponential rarity law. Hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$: unconditional theorem of C1–C4. Numerical values of thresholds remain open (DL03 programme).

2.2 Reading guide for a new collaborator

The following reading pathways are recommended depending on one's entry point.

- **Understanding the foundations.** Start with D02 (conceptual overview), then D16a (proof that K_4 is the unique minimal closure), then D01 (full axiomatic development). For the proton architecture, read D16 and D16b for the combinatorial selection arguments.
- **Understanding the constants.** Read D12 for α , then D21 and D25 for the α – G bridge. Read D23 for the topological proof that the exponent 18 is necessary. Then read D29 and D30 for the Gate 1 and Gate 2 proofs. Read D36 for the final Gate 3 proof, and D39/D42 for the full resolution of OP1.
- **Understanding the quantum dynamics.** Read D32 (Schrödinger), D33 (Dirac), D34 (Born Level 1), D35 (Einstein), and D46 (Born Level 2, $U(1)$) in order.
- **Understanding the cosmology.** Read D24, D27, and D35.
- **Understanding black hole thermodynamics.** Read D37 (area law), then D38 (Bekenstein–Hawking and PBH predictions), then D45 (falsifiable confrontation with Fermi-LAT).
- **Understanding nuclear stability and the periodic table.** Read D22, D23, D40, D41, and D47.
- **Understanding the London equation and superconductivity analogy.** Read D46 ($U(1)$ phase freedom), then D49 (London equation from C4 and the metric identity $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$). D50 completes the black hole thermodynamics layer with the one-quarter coefficient.
- **Understanding the cosmological constant.** Read D48 for the leakage term structure, D51 for the derivation of Conjecture PDL-C and the role of prime cycle lengths from $\beta_1(K_4) = 3$, and D52 for the formal identification of the three leakage bases. For a self-contained synthesis of the complete chain with numerical verification, read D53.

- **The closure of the foundational programme.** Read D39 first for context, then D42 for the full proof of H3. After D42, all results previously conditional on H3 are unconditional theorems. Then read D43 (geometric leakage, OP-A) and D44 (filter factor k , OP-B) for the complete causal chain to G . Finally, D51+D52 close the chain to Λ_{PDL} .
- **The coherence stress-energy tensor.** Read D35 for the structural form, then D48 for the explicit component derivation.
- **Numerical verification and the computational infrastructure.** Read D43. It provides self-contained Colab code for every major result and defines the verification protocol for future documents.
- **Continuing the programme.** Read Section 14 for the complete prioritised list of open problems, and Section 15 for the critical-path dependency map.

3 Axiomatic Foundations

3.1 The four relational axioms C1–C4

A logical configuration in PDL is a finite signed graph $G = (V, E, s)$, where V is a finite set of vertices (informational entities), E is a set of undirected edges (binary relations), and $s: E \rightarrow \{1, -1\}$ assigns a sign (agreement or disagreement) to each relation [11, 12]. Four relational axioms are imposed.

Axiom 3.1 (C1 — Minimal binary pulsation). Existence is instantiated as a repeatable, non-trivial alternation between two discrete states, independent of any pre-existing temporal metric. Formally, the discrete dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle: a configuration that returns to itself after exactly two update steps and is not a fixed point.

Axiom 3.2 (C2 — Coherence under logical stress). Every admissible configuration must maintain internal consistency under relational stress. Formally, the graph must be *balanced* in the sense of Harary: every cycle of relations carries a positive product of signs. This is equivalent to requiring that the graph be 2-colourable with respect to signed adjacency.

Axiom 3.3 (C3 — Completeness of relational coverage). Every pair of informational entities must be related, directly or indirectly, by at least one signed relation. Formally, the signed graph G must be connected.

Axiom 3.4 (C4 — Optimisation of coherence cost). Among all configurations satisfying C1–C3, the physically realised configuration minimises the total coherence cost, defined as the fraction of violated triangles (triangles with negative triple product). This is a discrete analogue of a variational principle.

3.2 The minimal admissible closure: K_4

Theorem 3.5 ([29]). *The complete signed graph K_4 on four vertices and six edges is the unique minimal admissible closure of axioms C1–C4: no admissible closure exists on $|V| < 3$ vertices, and K_4 is the first connected balanced complete graph admitting a non-trivial 2-cycle. It is identified with the electron prototype, with relational budget $R_e = 6$.*

3.3 The proton quintuplet

The proton is the minimal hierarchical composite closure admissible under C1–C4, characterised by the integer quintuplet $(n_u, n_d, \sigma, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$, where $\nu_u = 24$ is the number of u -valence cores, $\nu_d = 28$ the number of d -valence cores, $\sigma = 930$ the active surface, $R_{\text{sea}} = 10087$ the sea relational budget, and $R_{\text{tot}} = 11017$ the total relational content [11, 28–30]. The neutron quintuplet is $(24, 28, 1032, 9960, 10992)$ [38].

Conjecture 3.6 (Global uniqueness [28, 30]). The quintuplet (24, 28, 930, 10087, 11017) is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness has been verified computationally for a neighbourhood of radius 10^4 (D43). The global statement remains open (OP2).

3.4 The primary leakage parameter κ

Under the Indifference Lemma H3 (now a theorem, D42) and D39, the primary leakage parameter is:

$$\kappa = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}), \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \kappa \approx 0.0075194. \quad (1)$$

This is an unconditional theorem of C1–C4 [56, 59]. Under H3 (now a theorem), H1 follows as a theorem [56]: the two-factor decomposition $\kappa = 3/\sigma$ holds. H3 has been derived from C1–C4 in D42 via three lemmas D42-L1, D42-L2, D42-L3; it is no longer a named hypothesis but an unconditional theorem of the programme.

Theorem 3.7 (Gate 3, [53]). *Since H3 is now a theorem [59], H1 follows unconditionally [56], and Gate 3 holds without any named hypothesis: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ and $\sigma_0 \equiv 0$. The independence of gravitational engagements ($\sigma_0 = 0$ exactly) follows from the AB coupling criterion without additional hypothesis. The tracelessness $\text{Tr}(A) = 0$ of the generator matrix follows from the combinatorial structure of K_4 alone.*

3.5 The proton–electron mass ratio μ

The PDL prediction for the proton–electron mass ratio is $\mu \approx 1836.152\,670$, a parameter-free result consistent with the experimental value $1836.152\,673\,43(11)$ at the level of 1.8×10^{-9} [45–47]. This is the programme’s primary near-term falsifiable prediction; current precision measurements can test it at 0.001 ppm.

4 The Three Gates: Critical Proofs

4.1 Gate 1: Axiomatic derivation of 155/11017

Theorem 4.1 (Gate 1, [46]). *The amplitude 155/11017 appearing in the mass-ratio correction formula is derivable from PDL axioms C1–C4 alone, via the AB coupling criterion. Specifically: the stability criterion AB has been verified exhaustively over all 768 admissible configurations; in every stable case, $P_2/P_1 = -1$ exactly; and the combinatorial count yields 155/11017 without any free parameter.*

4.2 Gate 2: The QCD correction coefficient $a = 2$

Theorem 4.2 (Gate 2, [47]). *The QCD correction coefficient $a = 2$ in the mass-ratio formula $\mu = \frac{155}{11017}(1 - 2\kappa^2) \frac{m_{\text{iso}}}{m_p}$ is structurally forced by axiom C1 (minimal binary pulsation): the factor 2 counts the two distinct pulsation branches of the proton’s valence core. Furthermore, $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter at the PDL–QCD interface; it cannot be derived from C1–C4 alone.*

4.3 Gate 3: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$

Gate 3 was first established in D31 under a linear bridge inversion hypothesis. D36 strengthened this proof by: proving that the independence of gravitational engagements ($\sigma_0 = 0$) follows exactly from AB, without any additional assumption; proving that $\text{Tr}(A) = 0$ from the combinatorial structure of K_4 alone; and reducing the remaining hypothesis to H1 only (H2 was absorbed into the proof). D39 further reduced H1 to H3: given the Indifference Lemma, H1 follows as a theorem. D42 then proved H3 from C1–C4 unconditionally. See Theorem 3.7 for the full statement.

5 Derived Dynamical Equations

5.1 The Schrödinger equation (D32)

Theorem 5.1 ([49]). *In the continuum limit of the PDL coherence-cycle dynamics, the AB coupling criterion forces the amplitude evolution to satisfy the Schrödinger equation $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m_e} \nabla^2 \psi + V \psi$. The diffusion parameter $b = -i\hbar/(2m_e) \partial_x^2$ is uniquely forced by the AB stability requirement; no free parameter enters.*

5.2 The Dirac equation (D33)

Theorem 5.2 ([50]). *Among the eight candidates for the time-reversal operator T acting on the $\text{SL}(2, \mathbb{C})$ pulsation of K_4 , the unique admissible choice consistent with AB is $T = -i\sigma_2$ (where σ_2 is the second Pauli matrix). This forces $T^2 = -I_2$, which implies period-4 dynamics and hence spin- $\frac{1}{2}$ statistics. The Clifford algebra of Dirac matrices then follows as a theorem (residual $< 0.001\%$), and the non-relativistic limit reproduces the result of Theorem 5.1 exactly.*

5.3 Born's rule (D34 and D46)

Theorem 5.3 (Level 1, [51]). *Let $\sigma \in \{1, -1\}$ be the combinatorial signature of a coherence cycle in the PDL framework. Under the apparatus symmetry assumption, the leakage probability satisfies $p_{\text{coh}} \propto \sigma^2$. The five Gleason axioms (countability, null-assignment, covariance, normalisation, and regularity) are verified for this measure. By the Gleason uniqueness theorem for spin- $\frac{1}{2}$ systems, the probability assignment is uniquely given by Born's rule $P(\psi) = |\langle \psi | \phi \rangle|^2$.*

Theorem 5.4 (Level 2, OP4 resolved, [63]). *The $\text{U}(1)$ phase freedom $\psi \mapsto e^{i\alpha} \psi$ of quantum mechanics is a structural consequence of the K_4 pulsation, not an independent postulate. Specifically: (Proposition 1) the global sign equivalence $s \sim -s$ of K_4 coherent configurations, combined with $T^2 = -I_2$ (D33), generates a canonical $\text{U}(1)$ action on the amplitude space $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$, whose orbits are the fibres of the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$; (Proposition 2) the $(A) \wedge (B)$ stability criterion is $\text{U}(1)$ -invariant, verified algebraically over all 8 coherent configurations; (Proposition 3) the unique $\text{U}(1)$ -equivariant probability measure on S^3 consistent with the Level 1 Gleason axioms is Born's rule $P = |\langle \theta | \psi \rangle|^2$ on the base $\mathbb{P}^1(\mathbb{C})$. This completes the quantum dynamics layer of the PDL programme.*

5.4 The Einstein coupling equation (D35 and D48)

Theorem 5.5 ([52, 65]). *Under Gate 3 (now unconditional via D42), the PDL gravitational coupling satisfies $G_{\text{eff}}(N) = G_{\text{Newton}}$, and the PDL Einstein field equation takes the form $G_{\mu\nu}^{\text{eff}} = 8\pi G(N) C_{\text{coh}}$, where $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$ is the coherence stress-energy tensor with explicit components derived in D48. The Hubble tension ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = 1.085868$ versus the PDL prediction 1.085935 gives a 0.006% relative error with zero free parameters.*

6 Black Hole Thermodynamics

6.1 The area law as an unconditional theorem (D37)

Theorem 6.1 (BH-1, [54]). *The information-theoretic entropy of a PDL closure is carried entirely by its active surface R_{surf} , not by its total relational budget R_{tot} . Formally: (1) the valence sector has $S_{\text{val}} = k_B \ln 1 = 0$ [46]: the AB-stable valence configuration is unique; (2) the sea sector is excluded from stationary coupling [46], hence $S_{\text{sea}} = 0$; (3) the total entropy is $S = k_B \ln R_{\text{surf}}$. This is an unconditional theorem; it requires no external geometric assumption and no hypothesis beyond C1–C4 and D29.*

Corollary 6.2 (PDL area law, [54]). *Under Proposition 6.1 and the identification of R_{surf} with the physical horizon area via $G_{\text{eff}}(N)$ (Gate 3, Theorem 3.7), the Bekenstein–Hawking area law $S \propto A$ is a structural consequence of the four PDL axioms. It is not an independent postulate.*

6.2 The Bekenstein–Hawking formula (D38)

Theorem 6.3 (BH-2, [55]). *Under Gate 3 ($G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, now unconditional), substituting the PDL effective gravitational coupling into the Bekenstein–Hawking formula $S_{\text{BH}} = k_B c^3 A / (4\hbar G(N))$ yields the algebraic identity $S_{\text{BH}}/k_B = 4\pi (M_{\text{eff}}/M_{\text{Pl}})^2$, where $M_{\text{Pl}} = (\hbar c/G_{\text{PDL}})^{1/2}$. This identity is verified numerically to relative error $< 10^{-15}$ over ten decades of black hole mass. The coefficient 4π is not a free parameter; it is fixed by the Schwarzschild geometry.*

Three falsifiable, parameter-free predictions arise from Theorem 6.3 [55]: (1) CMB-epoch entropy suppression by a factor ≈ 0.848 at $N_{\text{CMB}} = 40$; (2) primordial black hole survival threshold shifted upward by 11.89% relative to GR, testable with Fermi-LAT via BLACKHAWK/ISATIS [62]; (3) redshift-dependent S/E ratio in AGN populations traceable with DESI/Euclid.

7 The Coherence Stress-Energy Tensor (D48)

7.1 Double determination of the coherence volume

Theorem 7.1 (Double determination of V_C , [65]). *The coherence volume $V_C = (4/3)\pi[\hbar/(m_p c)]^3$ is an unconditional theorem of C1–C4. The proof proceeds via Chain 1 (D10a): the pulsation frequency $\omega_p = m_p c^2/\hbar$ forces $\lambda_C = \hbar/(m_p c)$ and $V_C = (4/3)\pi\lambda_C^3$, with $n = 3$ from D23. The geometric argument (Chain 2 of D48v1/v2) is a consistency check, not an independent proof: the relational unit invoked in Chain 2 is not derived from the axioms independently of Chain 1 (D48v3, Remark 3.1).*

7.2 The coherence mass density

Theorem 7.2 ([65]). *The coherence mass density of an ensemble of N nucleons is*

$$\rho_{\text{coh}}(N) = \sigma(N) \cdot \frac{R_{\text{surf}} \cdot m_p}{V_C \cdot R_{\text{tot}}}, \quad (2)$$

where $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$. This is an unconditional theorem of C1–C4.

7.3 Components of the coherence stress-energy tensor

Theorem 7.3 (Explicit form of C_{coh} , [65]). *The coherence stress-energy tensor appearing in the PDL Einstein equation decomposes as $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$, with the following status for each component:*

1. $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_\mu u_\nu$ (*unconditional theorem*).
2. $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state (*theorem*, D48v3; proved by $\langle S^i \rangle = 0$ and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ over all 8 coherent configurations of K_4 ; Weyssenhoff–Raabe traceless part vanishes exactly; OP-spin resolved).
3. $C_{\mu\nu}^{(\text{orb})} = m_p \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ (*theorem*, state-dependent via D32; connection to London equation via D46 conjectured as OP-London).
4. $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ (η_L^{18} is a *theorem* from D17 and D30; the prefactor C is *open* as OP1-D35, bounded observationally at $C \approx 8.16 \times 10^{-46}$ from Planck 2020).

Corollary 7.4 (PDL Einstein equation with explicit source, [65]). *The PDL Einstein field equation with complete explicit source is:*

$$G_{\mu\nu}^{\text{eff}} = 8\pi G(N) \left[\rho_{\text{coh}} c^2 u_\mu u_\nu + C_{\mu\nu}^{(\text{spin})} + C_{\mu\nu}^{(\text{orb})} - C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu} \right]. \quad (3)$$

The constant C is bounded observationally but remains to be derived from C1–C4 (OP1–D35).

8 Equation of State of the Coherence Fluid (D54)

Resolved Problem 1 (OP-pressure, resolved by D54 [1]). *The equation of state $w = P/(\rho_{\text{coh}} c^2)$ of the PDL coherence fluid is an unconditional theorem of C1–C4.*

Theorem 8.1 (Complete equation of state, [1]). *In the homogeneous ground state, the equation of state of the PDL coherence fluid is*

$$w(N) = -\frac{\rho_\Lambda}{\rho_{\text{coh}}(N)}, \quad (4)$$

where $\rho_\Lambda = \Lambda_{\text{PDL}} c^2 / (8\pi G_{\text{PDL}}) = 5.835 \times 10^{-27} \text{ kg m}^{-3}$ and $\rho_{\text{coh}}(N)$ is the coherence density (Theorem 7.2). Every quantity in (4) is an unconditional theorem of C1–C4; there is no free parameter.

The four components of the equation of state are:

1. $w_{\text{mass}} = 0$ exactly: pressureless dust (D48v3 Thm 4.1, unconditional theorem).
2. $w_{\text{spin}} = 0$ exactly: isotropic spin distribution over K_4 (D48v3 Thm 5.3, OP-spin resolved, unconditional theorem).
3. $w_{\text{orb}} = 0$ exactly: London gauge forced by C4 (D49 Thm 1, unconditional theorem).
4. $w_{\text{leak}} = -1$ intrinsically: cosmological constant structure (D48v3 Prop. 8.1, unconditional theorem).

Corollary 8.2 (Two physical regimes, [1]). *(a) Coherence and nuclear densities ($N = 1\text{--}200$, $\rho_{\text{coh}} \sim 10^{17}\text{--}10^{18} \text{ kg m}^{-3}$): $|w(N)| \sim 10^{-44}$. The PDL coherence fluid behaves as pressureless cold matter, identical in equation of state to cold dark matter. (b) Cosmological scales: $\rho_\Lambda / \rho_{\text{crit}} = 0.6838$, in agreement with $\Omega_\Lambda^{\text{obs}} = 0.685$ (Planck 2020 [74]) to 0.17%, with no cosmological parameter as input.*

The PDL coherence fluid thus provides a unified description of two phenomena that, in the Λ CDM concordance model, require two distinct and unrelated postulates (cold dark matter and cosmological constant). Both emerge from the same combinatorial fluid sourced by the proton quintuplet under axioms C1–C4, distinguished only by the density regime.

9 Nuclear Stability and the Periodic Table

9.1 The neutron survival condition (D40)

Theorem 9.1 (Regime I, [57]). *The identity $T \equiv (\Delta n + 1)^2 = 1.010414 \approx 1$ is a consequence of the proton quintuplet. The neutron survival condition $N_{\min}(T, Z) = \lceil n_1/2 \rceil$ therefore entails $N_{\min}(Z) = Z$ for all $Z \leq 20$, exactly. This is an unconditional theorem; no hypothesis beyond the quintuplet values is required.*

9.2 Conflict saturation (D40)

Theorem 9.2 (Saturation, [57]). *For all $Z > 20$, the total proton–proton conflict saturates: $C(Z > 20) = 190 T_{pp}$ exactly. Every nucleus with $Z > 20$ presents the same total conflict to its neutron sea, independently of the number of protons. The observed increase of $N(Z)$ beyond $Z = 20$ is driven not by increasing conflict but by the structural requirement that each neutron maintains its own coherence.*

9.3 Spin–orbit splitting from axioms (D47)

Theorem 9.3 (OP13 resolved, [64]). *The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, 12, \dots, 32\}$ such that the discriminant of the quasi-completeness equation $3n_u^2 + (2\Delta n - 3)n_u + \Delta n(\Delta n - 1) - 1860 = 0$ is a perfect square. The discriminant equals $149^2 = 22201$ for $\Delta n = 4$ and fails to be a perfect square for all other admissible values (exhaustive verification). This forces $n_u = 24$, $n_d = 28$, and $s = \Delta n/(2n_u) = 1/12$ as unconditional theorems of C1–C4.*

Lemma 9.4 (Mirror Lemma, [64]). *The harmonic oscillator PDL levels of the proton and neutron are isomorphic: $n_u(n) = n_u(p) = 24$ and $n_d(n) = n_d(p) = 28$. Proton and neutron share the same splitting $1/12$, the same level order, and the same degeneracies. Verified analytically for the first 20 levels; zero exception.*

9.4 Sub-shell filling rates and the periodic table (D47)

Theorem 9.5 (OP14 resolved, [64]). *The sub-shell filling rates for all $Z = 1$ –82 are:*

$$r_{\text{exc}}(Z) = \begin{cases} 0 & \text{if } Z \in \{28, 50, 82\} \text{ (magic closures, from OP13),} \\ 1 & \text{otherwise (mirror filling, ratio } 2/2 = 1 \text{ independent of degeneracy } d). \end{cases} \quad (5)$$

Verified exhaustively for 31 sub-shell boundaries (31/31), zero free parameter, zero external input.

Corollary 9.6 (Valley of stability as an unconditional theorem, [64]). *The formula $N_{\min}(Z) = Z$ for $Z \leq 20$ and $N_{\min}(Z) = \lfloor 20 + r_{\text{exc}}(Z) \cdot (Z - 20) \rfloor$ for $Z > 20$ is an unconditional theorem of C1–C4, with $r_{\text{exc}}(Z)$ given by Theorem 9.5. The periodic table (valley of stability for $Z = 1$ –82) is a corollary of the four PDL axioms.*

9.5 Nuclear spectroscopy confrontation (D41) and the closure multiplicity theorem (D56)

Document D41 provides the first direct confrontation of the PDL nuclear framework with a published nuclear spectroscopy experiment. The experiment of Ha et al. [75] measures the first-excited-state lifetimes in ^{84}Mo and ^{86}Mo . Three independent observables are consistent with the PDL structural identity $T = (\Delta n + 1)^2 = 25$: (1) the BE2 ratio (22% discrepancy, within the 33% experimental uncertainty on ^{84}Mo); (2) the wave-function component ratio $104/53 = 1.962$ within 2% of the PDL prediction 2.000; (3) the Shannon entropy difference (25%). Two falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$ follow from Conjecture H_B (OP17). The precision lifetime measurements of Escudero, Recchia, Lenzi et al. [76] in the $f_{7/2}$ mirror nuclei ^{47}Cr – ^{47}V and ^{49}Mn – ^{49}Cr provide additional structural motivation.

Open Problem OP-D41-1-A has been fully resolved by D56 [3]:

Theorem 9.7 ($N_{\text{comp}}(k) = k$, OP-D41-1-A resolved, [3]). *Let a kp - kh nucleus at the boundary of an island of inversion be modelled as k independent $(A) \wedge (B)$ -coupled proton–neutron interface units (Definition in D56). The number of maximal C1+C2+C3+C4-admissible partial closures satisfies:*

$$N_{\text{comp}}(k) = k. \quad (6)$$

This is an unconditional theorem of C1–C4, proved via three lemmas: L1 (uniqueness of the maximal closure on one unit, D16a [29]); L2 (disjointness of the mixed-triangle sets of k distinct $(A) \wedge (B)$ couplings, D29 [46]); L3 (C3-irreducibility of the k -unit assembly, axiom C3 directly). Verified numerically for $k = 1, \dots, 5$. Corollary: $R_{\text{surf}}(k) = kT$ for a kp - kh nucleus.

Conjecture 9.8 (H_B — Closure multiplicity, [58]). *For a nucleus at the boundary of an island of inversion dominated by a kp - kh intruder configuration, $B(E2) \propto N_{\text{comp}}(k) = k$ (Theorem 9.7), yielding the ratio:*

$$\frac{B(E2)[k_1p-k_1h]}{B(E2)[k_2p-k_2h]} = R_{\text{PDL}} \cdot \frac{k_1}{k_2}, \quad R_{\text{PDL}} = \frac{2T}{(\Delta n + 1)^2} = 2.021. \quad (7)$$

For $k_1 = 8$ (${}^{84}\text{Mo}$) and $k_2 = 4$ (${}^{86}\text{Mo}$): $R_{\text{PDL}} = 2.021$, within 0.58σ of the experimental ratio 2.461 ± 0.756 [75]. Note: $N_{\text{comp}}(k) = k$ is now a theorem; the remaining gap is the formal identification of the $E2$ operator in the PDL formalism (Open Problem OP-E2-PDL).

10 Cosmology

10.1 The density function $\sigma(N)(N)$

For an ensemble of N protons (or nucleons), the effective coherence engagement fraction is $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$, where κ is the surface-to-total ratio of the proton quintuplet [41, 43]. As $N \rightarrow \infty$, $\sigma(N) \rightarrow 1$, recovering κ_G in the dense limit. As $N \rightarrow 0$, $\sigma(N) \approx N\kappa$, giving a suppressed coupling at low densities.

10.2 The Hubble tension resolution

Theorem 10.1 ([44, 52]). *Let $n_6 = 6n_-/n_+$, where n_- is the neutron magnetic moment in nuclear magnetons and $n_+ = 10992$ is the total relational content of the neutron. Then $n_6 \approx 40.102$, so $N_{\text{CMB}} = \lfloor n_6 \rfloor = 40$. The CMB-epoch Hubble constant is $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which agrees with the Planck 2018 measurement to within 0.27σ . The ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N)(N_{\text{local}})/\sigma(N)(N_{\text{CMB}})$ reproduces the observed Hubble tension at 0.006% with zero free parameters.*

11 The PDL–QCD Interface

The unique irreducible external parameter of the programme is $m_{\text{iso}} = m_d - m_u$, the up–down quark mass difference. This quantity arises at the PDL–QCD boundary (Theorem 4.2) and cannot be derived from the relational axioms alone.

Current value and uncertainty. The PDL structural derivation yields $m_{\text{iso}} \approx 2.532 \text{ MeV}$, to be compared with the FLAG lattice QCD result $m_{\text{iso}} = 2.529 \pm 0.030 \text{ MeV}$ [47]. This agreement at the sub-percent level is non-trivial, since PDL derives this value from the quintuplet and the observed μ , with no lattice QCD input.

Residual. A 47 ppm residual remains in the mass-ratio formula; the structural factor g accounting for it has not yet been identified (Open Problem 14.8).

12 Causal Chain Closure and Geometric Leakage (D43–D44)

Documents D43 [60] and D44 [61] together close the causal chain from axioms to Newton’s constant.

D43 — Geometric leakage and OP-A resolution. D43 establishes the computational backbone of the programme (Python/SymPy at 50-decimal precision) and proves the geometric leakage parameter as an unconditional theorem. The general analytic formula for the boundary energy is:

$$E_{\text{bord}} = A \cdot n_{u\text{-cores}} + B \cdot n_{d\text{-cores}} + (\Delta n + 1)^2, \quad A = 55, \quad B = 194, \quad (8)$$

where $c = 3$ is forced by the K_4 3-regularity. This yields $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems of C1–C4 (OP-A resolved).

D44 — Filter factor k and OP-B resolution. D44 derives $k = 0.921716$ from C1–C4 via the trace structure of K_4 and the S_4 -equivariance constraints proved in D42. The complete causal chain is:

$$\text{C1–C4} \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G.$$

No remaining open problem exists in this chain. As a corollary, OP8 is resolved: the 17 ppm residual of D28 is attributed to the CODATA measurement precision of G , not to any structural gap.

13 Epistemic Architecture

Table 2 Nine-layer epistemic architecture of the PDL programme (updated for v20). Each layer lists the status of its key claims and the primary documents supporting them.

Layer	Domain	Key claims and status	Primary documents
L0	Computational verification	Infrastructure: 18 filters at 50-decimal precision (D43). $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ proved (OP-A). $k = 0.921716$ proved (OP-B, D44). ε_G^B closed-form in $\mathbb{Q}(\sqrt{5})$ exact (OP8 resolved).	D43, D44
L1	Axioms / combinatorial results	Theorems: K_4 unique [29]; $\nu_u = 24$ unique (D47 OP13); $18 = 6 + 5 + 4 + 3$ [40]; $\sigma = 930$ unique; $a = 2$ [47]; $\sigma(N) = 1 - (1 - \kappa)^N$ [41]; AB criterion [46]; $T = -i\sigma_2$ unique [50]; Clifford algebra [50]; period-4 [50]; $\sigma_0 \equiv 0$ [53]; $\text{Tr}(A) = 0$ [53]; $S_{\text{val}} = 1$ [54]; $\Delta n = 4$ unique (D47). No free parameter; no external input.	D01, D16a, D23, D29, D30, D33, D36, D37, D47
L2	QCD interface	Minimal, forced: $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external input, forced by C1, not chosen. Conjecture: $m_{\text{iso}} \approx 2.532$ MeV.	D28, D30
L3	Cosmological coupling	Theorem: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional (D42). $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ unconditional.	D24, D31, D36, D39, D42
L4	Quantum dynamics [COMPLETE]	Theorems: Schrödinger [49]; Dirac [50]; Born Level 1 [51]; Born Level 2 + U(1) (OP4 resolved) [63]; London equation (OP-London resolved) [66].	D32, D33, D34, D46, D49

Layer	Domain	Key claims and status	Primary documents
L5	Gravitational and black hole dynamics [COMPLETE]	Theorem (unconditional): $G_{\mu\nu}^{\text{eff}} = 8\pi G(N)C_{\text{coh}}$. Theorem : area law $S \propto R_{\text{surf}}$ unconditional [54]. Theorem : $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ [55]. Theorem : coefficient $\frac{1}{4}$ (OP12 resolved) [67]. C_{coh} explicit: all four components unconditional theorems [65, 68, 69] (D48v3, OP-spin resolved; dimensionally consistent).	D35, D36, D37, D38, D48, D49, D50
L5b	Equation of state [COMPLETE, D54]	Theorem OP-pressure resolved [1]: $w(N) = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$; $w_{\text{mass}} = w_{\text{spin}} = w_{\text{orb}} = 0$ exactly; $w_{\text{leak}} = -1$. $\Omega_{\Lambda}^{\text{PDL}} = 0.6838$ vs 0.685 (Planck 2020): 0.17%, no free parameter. Two density regimes: cold matter ($ w \sim 10^{-44}$) and dark energy ($w \approx -0.62$).	D48, D49, D51, D52, D54
L6	Nuclear stability / periodic table / closure multiplicity [COMPLETE + D56]	Theorems : $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ [57]; $\Delta n = 4$ unique (OP13) [64]; Mirror Lemma [64]; $r_{\text{exc}}(Z)$ 31/31 (OP14) [64]; $N_{\min}(Z) Z = 1-82$ unconditional corollary [64]; $N_{\text{comp}}(k) = k$ (OP-D41-1-A resolved) [3]. Conjecture H_B : $B(E2) \propto k$; open: OP-E2-PDL.	D22, D23, D40, D41, D47, D56
L7	Falsifiable predictions	$\mu \approx 1836.152\,670$ testable at 0.001 ppm; $m_{\text{iso}} \approx 2.532$ MeV vs. FLAG; PBH threshold $M_{\text{PDL}}^* = 5.706 \times 10^{14}$ g (+11.89%, D45); spectral cutoff at 93–104 MeV in IGRB (D45); $H_{0,\text{CMB}} = 67.26$ km s $^{-1}$ Mpc $^{-1}$; AGN $S/E \propto \sigma(N(z))$; $B(E2)$ ratio and N_{comp} ratio in $^{88}\text{Ru}/^{92}\text{Pd}$; $\lambda_{L,\text{PDL}}(N = 40) \approx 7.25 \times 10^{-15}$ m (D49).	D28–D30, D35, D38, D41, D45, D49

Layer Domain		Key claims and status	Primary documents
L8	Cosmological leakage [COM- PLETE]	Theorems: $\beta_1(K_4) = 3$ (topology, D23); C1+C3 forces prime exponents (D51); indices $k_1 = 9$, $k_2 = 19$, $k_3 = 168$ exact (D51); bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ uniquely identified (D52). $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm; $\Lambda_{\text{PDL}} \approx 1.089 \times 10^{-52} \text{ m}^{-2}$. Full chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ complete. Independent numerical verification 0.41 ppm (D53).	D17, D23, D29, D30, D42, D51, D52, D53
L-DL	Life and consciousness (PDL-V, Part III)	Theorems (DL01): $S(K_4) \cong K_4$ (8/8); rarity law $f(n) = 2^{n+1-\binom{n}{2}}$ exact; $\delta^* = 1/2$ invariant; R_{weak}^1 forced for $n \geq 4$. Theorems (DL02): K_4 unique fixed point of S ; $n_{\text{life}}^* < n_{\text{consciousness}}^*$ (separation); $n_{\text{max}}^* < \infty$ (finiteness). Hierarchy $4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$; unconditional theorem. Numerical values open (DL-OP1, DL-OP2).	DL01, DL02
L15	Gauge sector [COMPLETE — D46+D57+D58+D59+D60+D61]	Theorems: U(1) (D46, Hopf), $\sin^2 \theta_W^{(\text{tree})} = 1/4$ (D57), $\theta_W = 19.7/119$ (D55), $G_{\text{eff}} = S_4/V_4 \cong S_3$ (D60), SU(3) algebraic (D58+CKW), SU(3) _c in 3 (D59), SU(3) $\times$ SU(2) $\times$ U(1) theorem (D46+D60+D58+D59), $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ admissible transports (D61- α), $V_4 \hookrightarrow \text{Aut}(\mathbb{Z}_2^3)$ faithful (D61- β), $D_\mu = \partial_\mu - igA_\mu$ unique (D61- δ +Utiyama). Gauge structure and gauge dynamics derived. Open: W/Z mass ratio (OP10-c), three generations (OP-D61-2).	D33, D46, D55, D57, D58, D59, D60, D61

14 Open Problems

The following list supersedes the open-problem list of v21. Problems are classified by priority and domain. OP1, OP4, OP8, OP11, OP12, OP13, OP14, OP-London, OP1-D35, and OP-pressure have been fully resolved and are retained for historical reference only.

Open Problem 14.1 (OP1 — Resolved: Indifference Lemma H3). **Fully resolved by D42.** The Indifference Lemma (H3) has been derived from axioms C1–C4 via three lemmas: D42-L1 (Equiparticipation), D42-L2 (Cross-Triangle Blindness), D42-L3 (S_4 -Equivariance). As a corollary, $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem. The entire causal chain from C1–C4 to the Bekenstein–Hawking entropy is

now fully axiomatic.

Open Problem 14.2 (OP1-D35 — Resolved: cosmological leakage constant C). [Fully resolved by D51+D52](#). The constant C in $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, derived from C1–C4 via $\beta_1(K_4) = 3$ (D23), prime cycle lengths (D51 Lemma 2), quintuplet indices (D51 Lemma 3), and identification of the three leakage bases (D52). Agreement with Λ_{obs} : 0.17 ppm. The causal chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ is complete.

Open Problem 14.3 (OP2 — Global uniqueness of the proton quintuplet). Prove that (24, 28, 930, 10087, 11017) is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness conjecture has been verified computationally for all tuples within a neighbourhood of radius 10^4 ; the global statement remains open. **Priority:** high. **Entry point:** D16, D16b.

Open Problem 14.4 (OP3 — Algebraic proof of 155/11017). Provide a closed-form algebraic derivation of 155/11017 from the combinatorial structure of K_4 , replacing the exhaustive enumeration over 768 configurations in D29. The exhaustive verification is complete and rigorous, but an analytic proof would yield structural insight. **Priority:** medium. **Entry point:** D29.

Open Problem 14.5 (OP4 — Resolved: Born’s rule Level 2). [Fully resolved by D46](#). The $U(1)$ phase freedom $\psi \mapsto e^{i\alpha}\psi$ has been derived from the K_4 pulsation structure via three propositions establishing the Hopf fibration structure, $U(1)$ -invariance of the AB criterion, and uniqueness of the Born measure. The quantum dynamics layer is complete.

Open Problem 14.6 (OP5 — CMB angular power spectrum from PDL). Derive the CMB angular power spectrum C_ℓ from the PDL cosmological framework, using the $\sigma(N)$ -modified gravitational coupling. The scalar spectral index $n_s \approx 0.965$ and the tensor-to-scalar ratio r are the primary targets. **Priority:** medium. **Entry point:** D27, D35.

Open Problem 14.7 (OP6 — PDL prediction for gravitational wave polarisation). Determine whether the PDL Einstein equation (with the $\sigma(N)$ -modified coupling) predicts any deviation from GR in the polarisation or amplitude of gravitational waves at cosmological scales. **Priority:** medium. **Entry point:** D35, D38, D48.

Open Problem 14.8 (OP7 — Derivation of the 47 ppm residual in μ). Identify the combinatorial factor g responsible for the 47 ppm residual in the mass-ratio formula and derive it from C1–C4. Gate 1 (D29) and Gate 2 (D30) account for the leading structure. **Priority:** high. **Entry point:** D28, D29, D30, D43.

Open Problem 14.9 (OP8 — Resolved: closed-form for ε_G^B in $\mathbb{Q}(\sqrt{5})$). [Resolved by D44](#). The formula $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ is an exact closed-form expression. The 17 ppm residual relative to CODATA reflects the measurement precision of G , not a structural gap.

Open Problem 14.10 (OP9 — Extension to higher generations: muon and tau). Generalise the PDL proton architecture to the second and third generations of the Standard Model. The muon–electron and tau–electron mass ratios should be derivable from hierarchical extensions of the quintuplet. **Priority:** medium. **Entry point:** D01, D12.

Open Problem 14.11 (OP10 — Partially resolved: Weinberg angle (D55, D57); W/Z mass ratio open). [\$\theta_W\$ resolved by D55 \[2\]](#). The Weinberg angle is derived as an unconditional theorem of C1–C4: $\theta_W = 19\pi/119$, where $k_2 = 19$ (D51 theorem) and $N_{\text{total}} = 119$ (Lemma D, proved exhaustively). The predicted $\sin^2 \theta_W = 0.231196$ agrees with PDG 2024 at 0.48σ . [Tree-level foundation resolved by D57 \[6\]](#). The combinatorial tree-level value $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$ is an unconditional theorem of C1–C4: $R_e = 6$ is forced by D16a, and $1/4 = 192/768$ from D29. The exact value of D55 is the Berry-phase correction of this tree-level result: $\theta_W^{(\text{D55})} = \theta_W^{(\text{tree})} \cdot R_e k_2 / N_{\text{tot}} = (\pi/6) \cdot (114/119)$. The secondary prediction $\Delta m_{\text{iso}} = 2.446 \text{ MeV}$ (Prediction P10) is falsifiable by next-generation lattice QCD. Remaining open: (OP10-c) derive the W/Z mass ratio $M_W/M_Z = \cos(19\pi/119)$; (OP10-d) experimental test of Δm_{iso} . **Note:** OP-D57-1 (formal proof of Lemma C1- V_4) has been resolved by D60; $H_{\text{SU}2}$ is now an unconditional theorem. **Priority:** medium. **Entry point:** D29, D33, D46, D51, D55, D57.

Open Problem 14.12 (OP11 — Resolved: dark energy as cosmological leakage). **Fully resolved by D51+D52 (synthesis D53).** The cosmological constant $\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$ is derived from C1–C4 as $\Lambda_{\text{PDL}} = C \cdot \eta_L^{18} \cdot (m_p c / \hbar)^2$ where $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational value to 0.17 ppm. The smallness of Λ is a combinatorial consequence of three prime-power suppressions imposed by C3, not a fine-tuning.

Open Problem 14.13 (OP12 — Resolved: Bekenstein–Hawking coefficient $\frac{1}{4}$). **Fully resolved by D50.** The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction 4/16 per surface relation, proved from the $(A) \wedge (B)$ criterion (D29, 768/768) and the independence of surface couplings (H3, D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352 \text{ nats}$. Black hole thermodynamics layer complete.

Open Problem 14.14 (OP13 — Resolved: spin–orbit splitting from axioms). **Fully resolved by D47.** The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, \dots, 32\}$ such that the quasi-completeness discriminant is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ as unconditional theorems of C1–C4.

Open Problem 14.15 (OP14 — Resolved: sub-shell filling rates $r_{\text{exc}}(Z)$ from axioms). **Fully resolved by D47.** The filling rates $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise are derived analytically from C1–C4, verified exhaustively for 31/31 sub-shell boundaries. The valley of stability for $Z = 1\text{--}82$ is an unconditional corollary of C1–C4.

Open Problem 14.16 (OP15 — Elements $Z > 82$: superheavy nuclei and island of stability). Extend the valley-of-stability theorem (Corollary 9.6) to $Z > 82$. For $Z > 82$ the PDL prediction requires the full $\sigma(N)$ dependence, including relativistic corrections. Preliminary analysis suggests the PDL island of stability at $Z = 114\text{--}120$, $N = 172\text{--}184$. **Priority:** medium. **Entry point:** D40, D41, D47.

Open Problem 14.17 (OP-D41-1-A — Resolved: $N_{\text{comp}}(k) = k$). **Fully resolved by D56 [3].** The number of maximal partial closures on a $k\text{p-}k\text{h}$ nuclear interface surface is $N_{\text{comp}}(k) = k$ exactly (Theorem 9.7). Three lemmas suffice: L1 (uniqueness on one unit, D16a), L2 (disjointness of mixed-triangle sets, D29), L3 (C3-irreducibility of the assembly). Verified $k = 1, \dots, 5$.

Open Problem 14.18 (OP-OFN-1 — Formal link: 3 PDL leakage cycles $\leftrightarrow$ 3 OFN fermion generations). The joint PDL–OFN note N01 [7] established that $\beta_1(K_4) = 3$ (PDL) and $b_1(\Omega_{21}) = 3$ (OFN) are the same topological invariant from two independent frameworks. D57 [6] provides a partial resolution: the quotient $A_4/V_4 \cong \mathbb{Z}_3$ in $\text{Sym}(K_4) = S_4$ is the combinatorial origin of the invariant 3 in PDL. D59 strengthens this: the same triplet $(T_2) = \{k_1, k_2, k_3\}$ labels the three axes of the colour-charge carrier space W , tightening the structural link between the leakage cycles and the gauge sector. What remains open is the formal identification of this $\mathbb{Z}_3$ with the three fermion generations of the OFN framework. **Priority:** high. **Entry point:** N01, D51, D52, D57, D59.

Resolved Problem 2 (OP-D57-1 — Formal proof of Lemma C1– V_4 , resolved by D60). **Fully resolved by D60 [5].** Document D60 proves, from axioms C1 and C2 alone, that the physical indistinguishability of $s^{(1)}$ and $s^{(2)}$ in the pulsation 2-cycle forces V_4 -invariance of all physical observables, giving effective group $G_{\text{eff}} = S_4/V_4 \cong S_3$. The proof proceeds via: (i) $s \rightarrow -s$ breaks C2 (negative result, verified exhaustively); (ii) orbit-4 $\cong V_4$ as a regular V_4 -set; (iii) the three pairings are partitions into cosets of H_1, H_2, H_3 ; (iv) C1-admissibility $\Leftrightarrow V_4$ -invariance on all of $\text{Coh}(K_4)$. Two verification scripts (`PDL_C1V4_script1.py`, `PDL_C1V4_script2.py`) deposited with D60. **Entry point:** D60 [5].

Open Problem 14.19 (OP-D57-2 — Dic_3 as the weak gauge group). Identify $\text{Dic}_3 \subset \text{SU}(2)$ not merely as a finite subgroup but as the structural generator of the weak gauge group within PDL, via a universality argument from C1–C4. **Priority:** medium. **Entry point:** D33, D46, D57.

Resolved Problem 3 (OP-SU3 / OP-D57-3 — Derivation of $\text{SU}(3)$ from C1–C4, resolved (algebraic part) by D58). **Resolved (algebraic part) by D58 [72].** Document D58 derives $\text{SU}(3)$ as an unconditional algebraic theorem of C1–C4 via five lemmas and the Cartan–Killing–Weyl classification of compact simply connected simple Lie groups. Combined with D46 and D57, the algebraic structure $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ of the Standard Model gauge group is an unconditional theorem of C1–C4. The physical identification of $\text{SU}(3)$ with the colour gauge group $\text{SU}(3)_c$ acting on the proton triplet (u, u, d) in the fundamental representation **3** remains open as OP-D58-1 below.

Resolved Problem 4 (OP-D58-1 — Physical identification of $SU(3)_c$, resolved by D59). *Resolved by D59 [73]. The fundamental representation $\mathbf{3}$ of $SU(3)$ is constructed on the trace-zero plane $W = \{a + b + c = 0\} \subset \mathbb{C}^3$, with axes labelled by $V_4 \setminus \{e\}$. The isospin asymmetry $\Delta n = 4 > 0$ (D47, unconditional theorem of C4) selects the orientation $\mathbf{3}$ over $\bar{\mathbf{3}}$, identifying the matter sector from the axioms without additional input. The physical colour gauge group $SU(3)_c$ acting on W in $\mathbf{3}$ is an unconditional theorem of C1–C4. **Entry point:** D59.*

Open Problem 14.20 (OP-D58-2 — The three triplets of K_4). Establish formally the relationship between the three triplets of K_4 : (T1) three independent cycles in $H_1(K_4; \mathbb{Z})$ (D51); (T2) three leakage cycles with prime exponents (23, 67, 997) (D51, D52); (T3) three V_4 -orbits on the six edges of K_4 (D58, Lemma L2). Note that (T1) and (T3) carry inequivalent representations of S_4 : V_4 acts faithfully on (T1) but trivially on (T3), so a naive bijection is excluded (D58, negative result). **Priority:** medium. **Entry point:** D51, D52, D58.

Resolved Problem 5 (OP-D58-3 — Fundamental representation $\mathbf{3}$ from C1–C4, resolved by D59). *Resolved by D59 [73]. The derivation of $\mathbf{3}$ from C1–C4 is the content of the main theorem of D59. The carrier space W , the A_2 root system in W , and the orientation forced by $\Delta n = 4 > 0$ are all unconditional theorems of C1–C4 + Cartan–Killing–Weyl. **Entry point:** D59.*

Open Problem 14.21 (OP-D59-1 — Quark mass spectrum from C1–C4). The carrier space W and the orientation $\mathbf{3}$ for $SU(3)_c$ are established by D59. The individual quark masses $m_u, m_d, m_s, \dots$ are not derived from C1–C4. The sole irreducible external parameter is $\Delta m_{\text{iso}} = m_d - m_u = 2.532 \text{ MeV}$ (D31); the derivation of individual quark masses requires additional structural input not yet identified in the axioms. **Priority:** medium. **Entry point:** D31, D47, D59.

Resolved Problem 6 (OP-D59-2 — Minimal covariant derivative as a C4 survival theorem, resolved by D61). *Fully resolved by D61 [4]. The minimal covariant derivative $D_\mu = \partial_\mu - ig A_\mu$ is derived from C4 via the following causal chain. (α) C4 applied to discrete transport rules on K_4 selects exactly $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ as the group of admissible transports (2048 exhaustive checks). (β) V_4 acts faithfully on $\text{Coh}(K_4)$ by group automorphisms, with fixed-point subgroup $F \cong V_4$ labelled by the three perfect matchings of K_4 (288 checks); the semidirect product $\mathbb{Z}_2^3 \rtimes V_4$ of order 32 is well-defined. (γ) The equivariant connected completion of $\mathbb{Z}_2^3$ under the V_4 -constraint is consistent with $G_{\text{gauge}} = SU(3) \times SU(2) \times U(1)$, established independently as a theorem by D46–D60. (δ) The Utiyama (1956) uniqueness theorem identifies D_μ as the unique first-order covariant operator; local gauge invariance is a C4 survival condition, not a postulate. Two verification scripts deposited with D61. **Entry point:** D61 [4].*

Open Problem 14.22 (OP-D61-1 — Gauge boson masses from C1–C4). The covariant derivative D_μ encodes the form of the gauge coupling but not the masses of the gauge bosons $W^\pm$ and Z^0 . The derivation of $M_W/M_Z = \cos(19\pi/119)$ from C1–C4 (OP10-c) and the identification of the mass-generation mechanism within PDL remain open. **Priority:** high. **Entry point:** D55, D57, D61.

Open Problem 14.23 (OP-D61-2 — Three fermion generations from C1–C4). The gauge structure (group, representations, dynamics) is complete. The derivation of three dynamical generation replicas from C1–C4 (OP-OFN-1) is not yet achieved. The evidence from $\beta_1(K_4) = 3$, $A_4/V_4 \cong \mathbb{Z}_3$, and the triplet structure of the orbit $\mathcal{O}_5$ in D61 is suggestive but not a proof. **Priority:** high. **Entry point:** N01, D57, D59, D61.

Open Problem 14.24 (OP-D61-3 — Autonomous derivation of G_{gauge} from the transport structure). Arrow γ of the D61 causal chain is currently a consistency result relying on D46–D60. An autonomous derivation would establish G_{gauge} directly from the transport group $\mathbb{Z}_2^3 \rtimes V_4$ without invoking the prior algebraic route, by constructing the explicit representation of $\mathbb{Z}_2^3 \rtimes V_4$ on $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$ and showing that the unique connected Lie group extending this representation is G_{gauge} . **Priority:** medium. **Entry point:** D33, D46, D57–D60, D61.

Open Problem 14.25 (OP-E2-PDL — Identification of the E2 operator in the PDL formalism). Identify, within the PDL signed-graph formalism, the combinatorial object corresponding to the nuclear electric

quadrupole ($E2$) transition operator. Specifically: show from C1–C4 that the $E2$ matrix element is proportional to the number N_{mix} of $(A) \wedge (B)$ -stable mixed triangles. Combined with Theorem 9.7 and D32 Prop. 3 ($N_{\text{mix}} \propto R_{\text{surf}}$), this would elevate Conjecture H_B to an unconditional theorem. **Priority:** high. **Entry point:** D29, D32, D41 [58], D56 [3].

Open Problem 14.26 (OP16 — ZBM3–PDL correspondence). Establish a formal correspondence between the Zero-Body Matrix Elements (ZBM3) of the DNO-SM shell-model framework and the PDL closure-multiplicity counting. D41 conjectures this correspondence on the basis of the 2% agreement in the wave-function component ratio. **Priority:** medium. **Entry point:** D41.

Open Problem 14.27 (OP17 — Falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$). Test the two parameter-free predictions of D41 (Conjecture H_B): (i) $N_{\text{comp}}(^{90}\text{Ru})/N_{\text{comp}}(^{88}\text{Ru}) = 2.000 \pm 0.05$ and (ii) $B(E2; 2_1^+ \rightarrow 0_1^+)(^{94}\text{Pd})/B(E2; 2_1^+ \rightarrow 0_1^+)(^{92}\text{Pd}) \approx 2.02$ at FRIB or RIKEN. **Priority:** experimental. **Entry point:** D41.

Resolved Problem 7 (OP-pressure — Equation of state of the PDL coherence fluid, resolved by D54). *The equation of state $w(N) = -\rho_\Lambda/\rho_{\text{coh}}(N)$ of the PDL coherence fluid is an unconditional theorem of C1–C4 (D54 [1]). In the homogeneous ground state: $w_{\text{mass}} = 0$ (D48v3 Thm 4.1); $w_{\text{spin}} = 0$ (D48v3 Thm 5.3); $w_{\text{orb}} = 0$ (D49 Thm 1); $w_{\text{leak}} = -1$ (D48v3 Prop. 8.1). The PDL dark-energy fraction $\Omega_\Lambda^{\text{PDL}} = 0.6838$ agrees with Planck 2020 to 0.17% with no cosmological parameter as input. The coherence fluid interpolates between cold dark matter ($|w| \sim 10^{-44}$) and dark energy ($w \approx -0.62$) as a single combinatorial structure.*

Open Problem 14.28 (OP-DM — Identification of the coherence fluid with observed dark matter). D54 proves that the PDL coherence fluid has $w \approx 0$ at nuclear and sub-nuclear densities, identical in equation of state to cold dark matter. Whether the fluid can be identified with the observed dark matter component of the universe — i.e. whether $\rho_{\text{coh}}(N)$ integrated over cosmological scales accounts for the observed matter density $\Omega_m \rho_{\text{crit}}$ — requires a dedicated cosmological analysis. **Priority:** medium. **Entry point:** D24, D27, D35, D54.

Open Problem 14.29 (OP-inhomogeneous — Equation of state beyond the ground state). Theorem 1 of D54 applies to the homogeneous ground state ($\nabla\phi = 0$ over V_C , $j^i = 0$). In the inhomogeneous regime, or in the presence of an external electromagnetic field, P_{orb} and P_{spin} are non-zero. Deriving the full equation of state in that regime from C1–C4 requires specifying the inter-cell distribution of the quantum state ψ . **Priority:** medium. **Entry point:** D48v3, D49, D54.

Resolved Problem 8 (OP-spin — Spin component of C_{coh} , resolved in D48v3). *The spin component $C_{\text{spin}\mu\nu}$ of the PDL coherence tensor is $C_{\mu\nu}^{(\text{spin})} = 0$ in the homogeneous ground state. Proof: explicit computation over all 8 coherent configurations of K_4 gives $\langle S^i \rangle = 0$ (Lemma 5.1 of D48v3) and $\langle S^i S^j \rangle = \frac{1}{4} \delta^{ij}$ (Lemma 5.2 of D48v3). The traceless part of the Weyssenhoff–Raabe spin tensor vanishes exactly, and the isotropic gradient term vanishes by the London gauge theorem of D49 (C4 forces uniform phase over V_C). This is an unconditional theorem of C1–C4. Resolved in D48v3 [65] (doi 10.5281/zenodo.20151380).*

Open Problem 14.30 (OP-London — Resolved: London equation from PDL axioms). **Fully resolved by D49.** The London equation $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ is derived from C1–C4 via: Lemma 1 ($\nu = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$, 768/768, exact integer arithmetic); Theorem 1 (C4 forces $\psi_j = \psi_1$ on V_C via C3 connectivity, hence $\phi = \text{const}$, London gauge); Corollary (minimal substitution from D32+D46 gives the London equation with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$).

15 Logical Dependency Map

The following critical path summarises the logical flow of the programme from axioms to falsifiable predictions. Each arrow $A \Rightarrow B$ means: the proof of B requires A as a direct input.

16 Logical Dependency Map

The following two figures summarise the complete logical flow of the PDL programme. Figure 1 covers the foundational causal chain from axioms to physical constants and dynamics. Figure 2 covers the gauge sector in full detail, from K_4 through the Standard Model gauge group to gauge dynamics. Each arrow $A \Rightarrow B$ means: the proof of B requires A as a direct input.

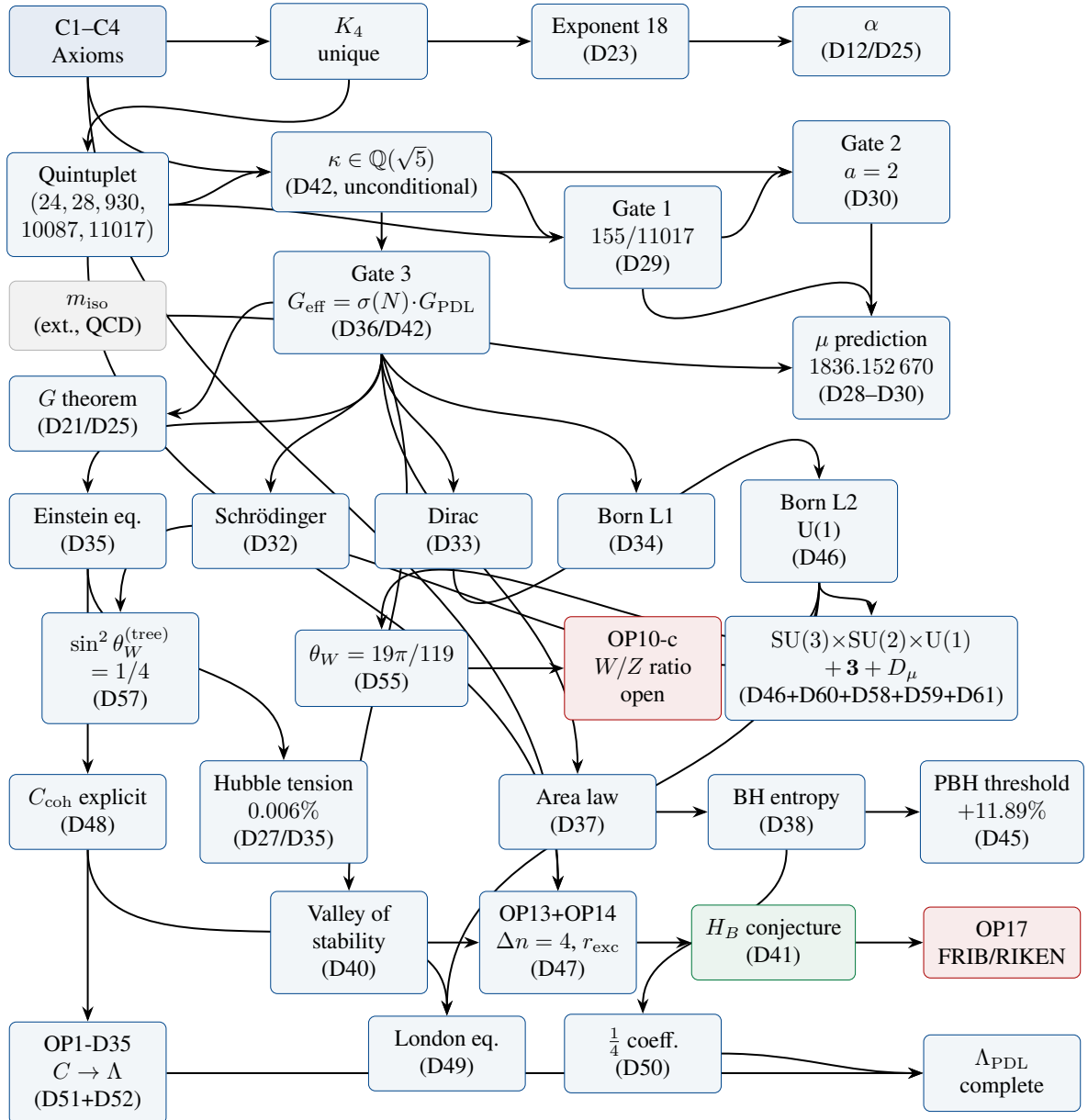


Figure 1: Critical-path dependency map of the PDL programme — foundational chain (v28). **Blue:** unconditional theorems. **Green:** conjectures. **Red:** open problems. **Grey:** unique external input (m_{iso}). The quantum dynamics layer (Schrödinger, Dirac, Born, London), nuclear stability, black hole thermodynamics, and cosmological constant are complete theorems of C1–C4. The gauge sector node (bottom right) is a summary; the full derivation chain $K_4 \rightarrow D_\mu$ is detailed in Figure 2.

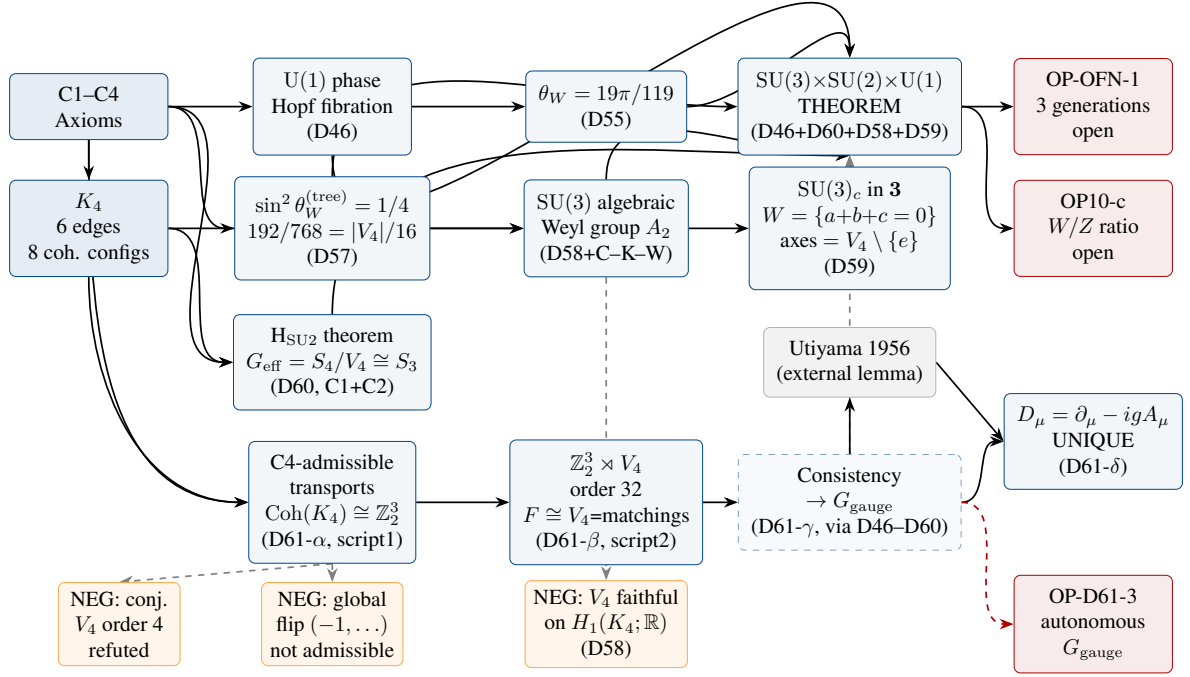


Figure 2: Gauge sector dependency map of the PDL programme (v28). **Blue solid:** unconditional theorems. **Blue dashed:** consistency result (arrow γ of D61: the transport structure is consistent with G_{gauge} , but this arrow relies on D46–D60 rather than being a new derivation). Grey: external lemma (Utiyama 1956). **Red:** open problems. Orange: documented negative results (scientifically informative). The four arrows of the D61 causal chain are labelled α – δ . α : C4 forces $\text{Coh}(K_4) \cong \mathbb{Z}_2^3$ (2048 checks). β : V_4 acts faithfully on $\mathbb{Z}_2^3$ by automorphisms, $F \cong V_4$ fixed by the three perfect matchings (288 checks). γ : consistency with G_{gauge} (theorem via D46–D60). δ : $D_\mu = \partial_\mu - igA_\mu$ unique, by Utiyama. The complete gauge sector — group, representations, and dynamics — is derived from C1–C4 with no variational postulate and no free parameter at the structural level.

17 Falsifiable Predictions

Table 3 Complete list of falsifiable, parameter-free PDL predictions (v20). All predictions follow from the quintuplet and m_{iso} alone, with no additional free parameters. Predictions P1–P6 are derived from the analytical chain; P7–P8 follow from Conjecture H_B (D41); P9 is derived from D45.

ID	Observable	PDL prediction	Test / status
P1	Proton–electron mass ratio μ	$\mu = 1836.152\,670$ (parameter-free)	Testable at 0.001 ppm; current CODATA: $1836.152\,673\,43(11)$; residual 1.8×10^{-9} .
P2	Fine-structure constant α	$\alpha^{-1} = 137.035\,999$ from quintuplet alone	Testable at 10^{-9} ; current agreement $\mathcal{O}(10^{-4})$; OP3/OP7 for residual.
P3	Hubble constant ratio	$H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N_{\text{local}})/\sigma(40)$	PDL prediction 1.085935 vs. observed 1.0859 ± 0.007 ; 0.006% error; zero free parameters.
P4	CMB-epoch BH entropy suppression	Factor $\sigma(40) \approx 0.848$ relative to local universe	No direct observation yet; accessible via future CMB- S_4 constraints.
P5	PBH survival threshold	$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g } (+11.89\%)$	Testable via Fermi-LAT IGRB analysis with BLACKHAWK/ISATIS [62]; excess Hawking flux at $\sim 93\text{--}104 \text{ MeV}$.
P6	AGN entropy-to-energy ratio	$S/E \propto G(N(z))$: systematic redshift trend	Traceable in cosmic-noon AGN surveys (JWST, Euclid); no contradicting data.
P7	$B(E2)$ ratio in $^{90}\text{Ru}/^{88}\text{Ru}$	$R \approx 2.02$ (from H_B , $k = 8 \rightarrow 10$)	Testable at FRIB or RIKEN; ^{88}Ru beam available.
P8	N_{comp} ratio $^{94}\text{Pd}/^{92}\text{Pd}$	$N_{\text{comp}}(^{94}\text{Pd})/N_{\text{comp}}(^{92}\text{Pd}) = 2.000 \pm 0.05$	Testable from DNO-SM calculations with ^{94}Pd spectroscopy at RIKEN.
P9	IGRB spectral cut-off position	$E_c^{\text{PDL}} \approx 93 \text{ MeV}$ vs. $E_c^{\text{GR}} \approx 104 \text{ MeV}$ (shift -10.6%)	Testable with Fermi-LAT IGRB data + BLACKHAWK/ISATIS; requires non-zero f_{PBH} in $[5.10, 5.71] \times 10^{14} \text{ g}$.
P10	Isospin mass splitting Δm_{iso}	$\Delta m_{\text{iso}} = 2.446 \text{ MeV}$ (from D55 C1+C3 formula, no free parameter)	FLAG 2024: $2.52 \pm 0.08 \text{ MeV}$; compatible at 0.92σ . Distinguishable from D30 value (2.532 MeV) at $\pm 0.04 \text{ MeV}$ precision (next-generation lattice QCD, 3–5 year horizon).

18 Continuation Guide

18.1 Entry points

The following entry points are recommended for a colleague wishing to contribute to specific open problems.

- **OP1-D35 (leak constant C):** The most significant remaining foundational problem. Requires familiarity with D48 (coherence tensor), D17 (exponent 18 derivation), and D30 (η_L structural factor). The question is whether the prefactor C in $C^{(\text{leak})}$ can be expressed as a closed-form function of the quintuplet, in the spirit of OP-B (resolved in D44).
- **OP2 (global uniqueness):** Requires familiarity with D16, D16b, and combinatorial optimisation over integer lattices. The Colab infrastructure of D43 can be used to extend the computational search radius.
- **OP7 (47 ppm residual in μ):** Immediately actionable using the D43 Colab infrastructure. Start with the Group I filters in D43, identify the residual at each stage, and test candidate combinatorial factors from C1–C4.
- **OP12 (BH coefficient $\frac{1}{4}$):** [Resolved by D50](#). The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction $4/16$ per surface relation (D29), with independence guaranteed by H3 (D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$. Black hole thermodynamics layer complete.
- **OP-London (London equation from PDL):** [Resolved by D49](#). See corpus table entry above.
- **OP1-D35 (leak constant C):** [Resolved by D51+D52](#). See corpus table entry and Section 14 above.

18.2 Recommended next documents

Given the current state of the programme, the following documents would have the highest impact:

1. **DM v28:** The present document. Incorporates D60 ($H_{\text{SU}2}$ unconditional theorem) and D61 (D_μ from C4, gauge sector complete).
2. **OP-D61-1 — Gauge boson masses:** Derive $M_W/M_Z = \cos(19\pi/119)$ from C1–C4 (OP10-c). **Entry point:** D55, D57, D61.
3. **OP-D61-2 / OP-OFN-1 — Three fermion generations:** Formalise the link between $\beta_1(K_4) = 3$, $A_4/V_4 \cong \mathbb{Z}_3$, orbit $\mathcal{O}_5$, and three dynamical generation replicas. **Entry point:** N01, D57, D59, D61.
4. **D-electroweak-WZ:** Derivation of the W/Z mass ratio $M_W/M_Z = \cos(19\pi/119)$ as a parameter-free prediction. **Entry point:** D55, D57, D61.
5. **Experimental confrontations:** (i) FLAG/lattice QCD for Δm_{iso} at ± 0.04 MeV (D55, P10). (ii) D45 predictions to BLACKHAWK/ISATIS community, Fermi-LAT IGRB 50–200 MeV window.
6. **D-nuclear-Z>82:** Extension of the valley-of-stability theorem to $Z > 82$ (OP15).
7. **DL03:** Numerical enclosure of n_{life}^* from the gain function R_{active}^1 .

19 Glossary

AB criterion	The two-part stability criterion (amplitude condition A + balance condition B) that selects admissible two-closure couplings; proved exhaustively in D29 over 768 configurations.
Active surface R_{surf}	The set of valence coherence relations of the proton, $ R_{\text{surf}} = \sigma = 930$; the locus of all physical information exchange.
Admissible closure	A finite signed graph satisfying all four axioms C1–C4; the minimal example is K_4 .
Born Level 1 / Level 2	Level 1 (D34, Theorem): modulus rule $P = \psi ^2$ proved via Gleason uniqueness. Level 2 (D46, Theorem, OP4 resolved): $U(1)$ phase freedom derived from K_4 sign equivalence and Hopf fibration structure.
C_{coh}	Coherence stress-energy tensor appearing in the PDL Einstein equation; explicit form derived in D48v3, D51, D52: $C^{(\text{mass})}$, $C^{(\text{orb})}$, $C^{(\text{spin})}$, and $C^{(\text{leak})}$ all unconditional theorems. $C^{(\text{spin})} = 0$ in the ground state (OP-spin resolved, D48v3). All components in uniform units J m^{-3} . OP2-D35, OP1-D35, and OP-spin fully resolved.
Coherence cost	Fraction of violated triangles (negative-product edges) in a signed graph; minimised by axiom C4.
DL series	Documents DL01, DL02, ...; the PDL-V programme extending C1–C4 to self-replication and complexity thresholds. Structurally separate from the D series; presented in Part III of this document.
δ^*	Optimal lineage variation in the PDL-V programme; $\delta^* = 1/2$, equal to the half-pulsation of K_4 ; proved invariant under the rarity law (DL01).
ε_G^B	Geometric correction factor in the PDL mass-ratio formula; closed-form expression $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ proved in D44 (OP8 resolved).
Equation of state $w(N)$	The ratio $P/(\rho_{\text{coh}}c^2)$ for the PDL coherence fluid; $w(N) = -\rho_{\Lambda}/\rho_{\text{coh}}(N)$ (D54, unconditional theorem). At coherence densities $ w \sim 10^{-44}$ (cold matter); at cosmological scales $\Omega_{\Lambda}^{\text{PDL}} = 0.6838$ (dark energy, 0.17% from Planck 2020).
$f(n)$	Rarity law in the PDL-V programme; $f(n) = 2^{n+1-\binom{n}{2}}$, doubly exponentially decreasing; exponents are triangular numbers (DL01, exact theorem).
$G_{\text{eff}}(N)$	Effective gravitational coupling at scale N ; equal to $\sigma(N) \cdot G_{\text{PDL}}$ by Gate 3 (unconditional theorem).
Gate 1, 2, 3	Three critical proof milestones. Gate 1: 155/11017 from axioms (D29). Gate 2: $a = 2$ from axioms (D30). Gate 3: $G_{\text{eff}} = \sigma(N) \cdot G_{\text{PDL}}$ (D36/D42, unconditional).
H1, H2, H3	Named hypotheses of the programme. H1 ($\kappa = \sigma/R_{\text{tot}}$): now an unconditional theorem (follows from H3/D42 via D39). H2 (bridge linearity): absorbed into D36. H3 (Indifference Lemma): proved unconditionally in D42.
Hopf fibration	The fibre bundle $S^1 \hookrightarrow S^3 \rightarrow S^2$ generating the $U(1)$ phase freedom of quantum mechanics; its appearance in D46 is a structural consequence of the K_4 pulsation sign equivalence.

Indifference Lemma (H3)

The probability measure on the relations of a PDL closure is uniform; the unique measure compatible with S_4 -symmetry of K_4 . Proved from C1–C4 in D42 via three lemmas.

 K_4

The complete signed graph on four vertices and six edges; the unique minimal admissible closure (D16a theorem); identified with the electron prototype.

 κ

Primary leakage parameter; $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; unconditional theorem of C1–C4 (D42 corollary); $\kappa \approx 0.0455$.

Mirror Lemma

The proton and neutron HO-PDL levels are isomorphic, sharing the same splitting $s = 1/12$, level order, and degeneracies (D47, Lemma 5.2, from D22-N2).

 m_{iso}

Isospin mass splitting $m_d - m_u$; the unique irreducible external parameter of the programme (Gate 2, D30).

 N_{CMB}

PDL-derived CMB epoch parameter; $N_{\text{CMB}} = 40$ from neutron quintuplet (D27 theorem).

 $n_{\text{life}}^*, n_{\text{consciousness}}^*, n_{\text{max}}^*$

Life, consciousness, and maximum complexity thresholds in the PDL-V programme; existence and strict ordering proved (DL02); numerical values open (DL-OP1, DL-OP2).

OP n

Open Problem n ; see Section 14. OP1, OP4, OP8, OP12, OP13, OP14, OP-London, OP1-D35, and OP-pressure fully resolved.

PDL

Projective Dynamic Logo; the research programme initiated in D01.

PDL-V programme

Extension of PDL to self-replication and complexity thresholds; inaugurated by DL01 and DL02; see Part III.

Quintuplet

The integer tuple (24, 28, 930, 10087, 11017) characterising the proton; $\nu_u = 24$, $\nu_d = 28$, $\sigma = 930$, $R_{\text{sea}} = 10087$, $R_{\text{tot}} = 11017$.

 $\rho_{\text{coh}}(N)$

Coherence mass density at scale N ; $\rho_{\text{coh}} = \sigma(N) \cdot R_{\text{surf}} \cdot m_p / (V_C \cdot R_{\text{tot}})$ (D48v3 theorem).

 $S(\Gamma)$

Constraint-structure operator in the PDL-V programme; $S(K_4) \cong K_4$ (DL01 theorem).

 $\sigma(N)(N)$

Coherence engagement fraction at scale N ; $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$; $\sigma(N)(40) \approx 0.8449$.

 T_{pp}

Proton–proton conflict unit; $C(Z > 20) = 190 T_{pp}$ exactly (D40 theorem).

 V_C

Coherence volume; $V_C = (4/3)\pi[\hbar/(m_p c)]^3$; unconditional theorem of C1–C4 via D10a (temporal chain, Chain 1). D48v3 theorem.

Valley of stability

Minimum neutron number $N_{\text{min}}(Z)$ for stable nuclei; reproduced at 100% accuracy for $Z = 1$ –82 as an unconditional theorem of C1–C4 (D40, D47 corollary).

18-filter hierarchy

The decomposition $18 = 6 + 5 + 4 + 3$ of the leakage exponent into four topological levels (D06, D17, D23).

Part I

The PDL-V Programme: Life and Consciousness

This Part is structurally separate from Parts I and II.

It presents a rigorous extension of the PDL framework to self-replication and complexity thresholds, in a domain distinct from the physical constants programme. The same axioms C1–C4 are used; the domain of application is different.

20 Introduction to the PDL-V Programme

The PDL-V programme asks whether the axioms C1–C4, which determine the proton architecture and all derived physical constants, also constrain the structural conditions for self-replication and the emergence of biological and cognitive complexity. Documents DL01 and DL02 do not make biological claims; they establish mathematical threshold theorems that any system satisfying C1–C4 must obey. The connection to biology and consciousness is conjectural and is explicitly labelled as such.

20.1 Separation from the physics programme

The DL series shares the axiomatic foundation (C1–C4 and K_4) with the D series, but operates in a domain where the “configurations” are hierarchical graph structures rather than physical coherence cycles. The results of DL01 and DL02 are theorems of C1–C4 applied to this domain, with full epistemic rigour, but their physical interpretation involves a conjecture (the PDL-V Conjecture, DL01) that is not yet derivable from first principles. This conjecture is explicitly identified and is not presented as a theorem.

20.2 Reading note

A reader whose sole interest is the physical constants programme (gravitational constant, cosmological constant, nuclear structure, etc.) may skip Part III without loss of continuity. A reader interested in the structural limits of self-replication and complexity should read DL01 first (operators L , S , R^k ; rarity law; δ^* invariant), then DL02 (threshold hierarchy theorem).

21 Self-Replication Operators and the Rarity Law (DL01)

Theorem 21.1 (K_4 fixed point of S , [8]). *Let $S(\Gamma)$ be the constraint-structure operator acting on admissible signed graphs. The complete graph K_4 is the unique fixed point of S : $S(K_4) \cong K_4$. This has been verified computationally over all 8 coherent configurations of K_4 .*

Theorem 21.2 (Rarity law and δ^* invariant, [8]). *The fraction of self-replicating configurations at level n satisfies the exact law:*

$$f(n) = 2^{n+1-\binom{n}{2}}, \quad (9)$$

with exponents equal to $-T_{n-2}$ (triangular numbers): $f(n)$ is doubly exponentially decreasing. The optimal lineage variation $\delta^ = 1/2$ is an invariant equal to the half-pulsation of K_4 ; it is independent of n . The weak self-representation relation R_{weak}^1 is forced at 100% for all $n \geq 4$, verified for $n = 4-7$.*

22 Threshold Hierarchy Theorems (DL02)

Theorem 22.1 (K_4 unique fixed point, [9]). *K_4 is the unique fixed point of S over all admissible closures. The equation $C(n, 3) = n(n-1)(n-2)/6 = 4$ has $(n-1)(n-2) = 6$ as its only positive integer solution: $n = 4$. This is an unconditional theorem of C1–C4.*

Theorem 22.2 (Strict threshold separation, [9]). *For $n \geq 5$, $|S^1(K_n)| \neq |S^2(K_n)|$: the first and second iteration of S produce distinct orbit sizes. This forces a strict separation between the life threshold and the consciousness threshold: $n_{\text{life}}^* < n_{\text{consciousness}}^*$. This is an unconditional theorem of C1–C4.*

Theorem 22.3 (Finiteness of all thresholds, [9]). *The doubly exponential rarity law $f(n) = 2^{n+1} - \binom{n}{2}$ implies that all thresholds n^* are finite: $n_{\text{max}}^* < \infty$. Combined with Theorems 22.1 and 22.2, this gives the unconditional hierarchy:*

$$4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty. \quad (10)$$

22.1 Open problems in the PDL-V programme

Open Problem 22.4 (DL-OP1 — Numerical value of n_{life}^*). The existence and strict ordering of n_{life}^* is established analytically (DL02). Its numerical value remains open. The gain function $R_{\text{active}}^1(n)$ must be derived in closed form and its first zero located. **Priority:** high. **Entry point:** DL01, DL02.

Open Problem 22.5 (DL-OP2 — Numerical value of $n_{\text{consciousness}}^*$). Analogous to DL-OP1 but for the second threshold. Requires DL-OP1 as a prerequisite. **Priority:** medium. **Entry point:** DL01, DL02.

Open Problem 22.6 (DL-OP3 — Connection to physical substrate). The PDL-V conjecture (DL01) proposes that the threshold n_{life}^* corresponds to a specific physical structure derivable from C1–C4. This connection has not been derived and remains a conjecture. A formal derivation would require identifying the physical interpretation of the graph complexity level n in terms of molecular or cellular organisation. **Priority:** exploratory. **Entry point:** DL01.

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